

Magnetic Field Models for Multipole Magnets in Cylindrical Coordinates

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1 Field expansion in multipoles

In this note, the field-point coordinates are r, θ, z and source-point coordinates are a, ϕ, z' . We also define unit vectors $\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ and $\hat{a} = \cos \phi \hat{x} + \sin \phi \hat{y}$. Then, the field point is $\mathbf{r} = r\hat{r} + z\hat{z}$ and the source point is $\mathbf{r}' = a\hat{a} + z'\hat{z}$. In general, except for a solenoidal component, the magnetic scalar potential in cylindrical coordinates for the magnetic field in the vacuum bore of a magnet can be represented by Fourier series in the azimuthal angle θ with coefficients that are functions of r and z :

$$V(r, \theta, z) = \sum_{m=1}^{\infty} [U_m^s(r, z) \sin m\theta + U_m^c(r, z) \cos m\theta] \quad (1)$$

In the above expression, the axisymmetric (solenoidal) term $m = 0$ has been omitted because it requires special treatment. The sine terms are the so-called normal-multipole terms and the cosine terms the skew-multipole terms*. In accelerator applications, z is the nominal beam axis.

The magnetic field vector is given by the gradient of the scalar potential:

$$B_r = \frac{\partial V}{\partial r} \quad B_\theta = \frac{1}{r} \frac{\partial V}{\partial \theta} \quad B_z = \frac{\partial V}{\partial z} \quad (2)$$

Both U_m^s and U_m^c satisfy the reduced Laplace equation :

$$\frac{\partial^2 U_m}{\partial r^2} + \frac{1}{r} \frac{\partial U_m}{\partial r} + \frac{\partial^2 U_m}{\partial z^2} - \frac{m^2}{r^2} U_m = 0 \quad (3)$$

Therefore U_m^s and U_m^c are the same except for multiplicative constants related to the strengths of the $\sin m\theta$ and $\cos m\theta$ components. The function U_m is called the reduced scalar potential. In analyzing field data, we can extract the normal multipole components $b_r^m(z)$, $b_\theta^m(z)$, and $b_z^m(z)$ by the usual Fourier integrals:

$$\begin{aligned} b_r^m(z) &= \frac{1}{\pi} \int_0^{2\pi} B_r(a, \theta, z) \sin m\theta d\theta & b_\theta^m(z) &= \frac{1}{\pi} \int_0^{2\pi} B_\theta(a, \theta, z) \cos m\theta d\theta \\ b_z^m(z) &= \frac{1}{\pi} \int_0^{2\pi} B_z(a, \theta, z) \sin m\theta d\theta \end{aligned} \quad (4)$$

In the the following, for simplicity, only the normal components will be retained and the superscript dropped; putting the skew components back in is straightforward. For a pure normal multipole component we write $V_m(r, \theta, z) = U_m(r, z) \sin m\theta$.

*Strictly speaking, the term "multipole" is properly used in the following two contexts: The first (not relevant here) is an expansion in spherical harmonics of the potential in the space outside of an extended charge or current distribution centered at the origin. The second is an expansion of a 2-D magnetic field in powers of the complex variable $(x + iy)$. The 2-D expansion is valid for the middle part of magnets long in z , or for the z -integrated 3-D field. In real accelerator multipole magnets, the poles in many cases are physically iron pole pieces, and there are $2m$ poles, with two poles for dipoles ($m = 1$), four poles for quadrupoles ($m = 2$), etc. My use is really a third one, where it refers to a 3-D potential in cylindrical coordinates with pure $\sin m\theta$ symmetry.

2 Derivation of the scalar potential for a sinusoidally-varying charge density

We note that since apparently no real magnetic monopoles exist in nature (at least as sources of the magnetic fields of concern here), the magnetic charges that appear in this report will always be fictitious, and will be used only in application of the Helmholtz theorem to field computation.

We start with a ring of radius a with a line-charge density $\lambda = \lambda_m \sin m\phi$ at $z = z'$. Then the open-boundaries scalar potential is

$$V_m^\lambda(r, \theta, z) = \frac{\lambda_m a}{4\pi} \int_0^{2\pi} \frac{\sin m\phi d\phi}{R(r, \theta, z; a, \phi, z')}, \quad (5)$$

where R is the distance between the field point r, θ, z and the source point a, ϕ, z' . Using $R = [r^2 + a^2 - 2ar \cos(\phi - \theta) + (z - z')^2]^{1/2}$, and defining the new variable $\alpha = \phi - \theta$, we now have

$$V_m^\lambda(r, \theta, z) = \frac{\lambda_m a}{4\pi} \left[\sin m\theta \int_{-\theta}^{2\pi-\theta} \frac{\cos m\alpha d\alpha}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{1/2}} + \right. \quad (6)$$

$$\left. \cos m\theta \int_{-\theta}^{2\pi-\theta} \frac{\sin m\alpha d\alpha}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{1/2}} \right] \quad (7)$$

$$= \frac{\lambda_m a}{2\pi} \sin m\theta \int_0^\pi \frac{\cos m\alpha d\alpha}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{1/2}} \quad (8)$$

The $\sin m\alpha$ term is zero because the integral of an odd trigonometric function over a complete period is zero. If the charge density $\sigma_m(z')$ is distributed in z' , the scalar potential is given by a double integral:

$$V_m^\sigma(r, \theta, z) = \frac{a}{2\pi} \sin m\theta \int_0^\pi \cos m\alpha d\alpha \int_{-\infty}^{+\infty} \frac{\sigma_m(z') dz'}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{1/2}} \quad (9)$$

In general, it is impossible to evaluate the double integral of Eq. 9 in closed form. For rapid evaluation of the scalar potential, field components, and the vector potential when the source distribution is represented by piecewise-continuous polynomials (splines in z' , etc.), it is possible to first integrate in z' over source distributions represented by piecewise-continuous polynomials (splines in z' , etc.), and then perform the integral of each piece over α by Bartky quadrature [10]. However, a detailed description of this approach is beyond the scope of this memorandum, which is mainly concerned with evaluating the on-axis gradient and its derivatives for input to particle beam-optics programs (see the next section). Alternatives to the Bartky quadrature approach for off-axis fields of extended distributions are to numerically integrate the charge-ring Legendre-function expressions developed in the following in z' or to numerically evaluate Fourier integrals involving modified Bessel functions.

Returning to the sinusoidal charge ring potential and introducing the variable p as defined by

$$p = \frac{a^2 + r^2 + (z - z')^2}{2ar} \quad (10)$$

we can write

$$V_m^\lambda(r, \theta, z) = \frac{\lambda_m a}{2\pi} \frac{1}{\sqrt{2ar}} \sin m\theta \int_0^\pi \frac{\cos m\alpha d\alpha}{(p - \cos \alpha)^{1/2}} \quad (11)$$

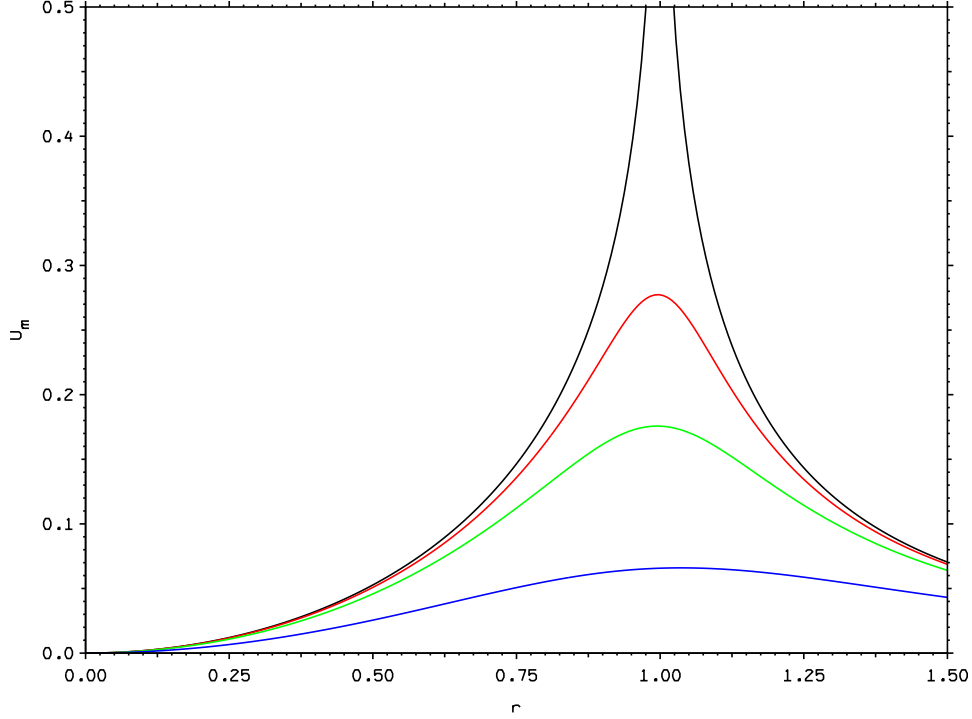


Figure 1: Plot of U_m vs. r for a charge ring at $z' = 0$ with $m = 2$ and $a = 1$ for various values of z . Black: $z = 0$; red: $z = 0.1$; green: $z = 0.2$; blue: $z = 0.5$.

The integral in Eq. 8 is not usually found in tables of integrals, but it can be found in Ref. [2], p. 1304, with n in place of m and $\cosh \mu$ in place of p . It can also be found in Ref. 3. Vol. 1, p. 156, Eq. 10 as a special case of an integral expression for Q_m^μ , with $\mu = 0$ and $\nu = m - 1/2$. Finally, the integral can be found in Ref. [6], p. 73. The references give

$$\int_0^\pi \frac{\cos m\alpha d\alpha}{(p - \cos \alpha)^{1/2}} = \sqrt{2} Q_{m-1/2}(p) \quad (12)$$

Putting this all together, we get the Legendre-function expression for the reduced scalar potential for a sinusoidally varying linear charge density $\lambda(\phi) = \lambda_m \sin m\phi$ on a ring of radius a at $z = z'$:

$$U_m(r, z) = \frac{\lambda_m a}{2\pi} \frac{1}{\sqrt{ar}} Q_{m-1/2}(p) = \frac{\lambda_m a}{2\pi} \frac{1}{\sqrt{ar}} Q_{m-1/2} \left[\frac{a^2 + r^2 + (z - z')^2}{2ar} \right] \quad (13)$$

Figure 1 is a plot of $U_m(r, z)$ vs. r for $a = 1.0$, $z' = 0$, $m = 2$ and various values of z for a charge ring with $\lambda_m = 1$. With the sign convention of Eqs. 2 (i.e. the field is the positive gradient of the scalar potential), a positive λ_m gives a positive B_r at the center of the first pole of a charge-sheet normal multipole magnet. A more complete description of the Legendre function $Q_{m-1/2}$ and its properties is given in Appendix A.

3 On-axis generalized gradient for the scalar potential for a ring with a sinusoidally varying line-charge density

The on-axis gradient, defined in this section, and its z derivatives serves as the interface of various magnet models with numerical map integration routines of high-order beam-optics codes, such as Marylie. For this purpose, a subroutine package that can compute the the gradient and its derivatives at arbitrary z values is required. The use of the on-axis generalized gradient in expansion of the magnetic field and potentials in cylindrical coordinates is given in Appendix B.

One of the elemental building blocks in the gradient subroutines is the sinusoidal charge ring. The other essential building block is the sinusoidal radially-pointing dipole ring, described in a later section.

The Legendre function Q_ν can be represented by Gauss's hypergeometric series in the variable $1/p^2$ (see Ref. [3], p. 122 or Ref. [5] p. 1016 **8.820** 2.)

$$Q_\nu(p) = \frac{\Gamma(\nu+1)\Gamma(1/2)}{2^{\nu+1}\Gamma(\nu+3/2)} \left(\frac{1}{p}\right)^{\nu+1} F\left(\frac{\nu}{2} + \frac{1}{2}, \frac{\nu}{2} + 1, \nu + \frac{3}{2}; \frac{1}{p^2}\right) \quad (14)$$

With $\nu = m - 1/2$ and $m > 0$ this becomes

$$\begin{aligned} Q_{m-1/2}(p) &= \frac{(2m-1)!!\pi}{2^{2m+1/2}m!} \left(\frac{1}{p}\right)^{m+1/2} F\left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; \frac{1}{p^2}\right) \\ &= \frac{\pi(2m-1)!!(2ar)^m\sqrt{ar}}{2^{2m}m![a^2 + r^2 + (z-z')^2]^{m+1/2}} F\left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; \frac{1}{p^2}\right) \end{aligned} \quad (15)$$

and Eq. 13 becomes

$$U_m(r, z) = \lambda_m a r^m \frac{(2m-1)!!a^m}{2^{m+1}m![a^2 + r^2 + (z-z')^2]^{m+1/2}} F\left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; \frac{1}{p^2}\right) \quad (16)$$

We note that the apparent singularity at $r = 0$ due to the $1/\sqrt{ar}$ factor in Eq. 13 is removed when U_m is expressed in hypergeometric form. For the special case $m = 0$ we have

$$U_0(r, z) = \lambda_m a \frac{1}{2[a^2 + r^2 + (z-z')^2]^{1/2}} F\left(\frac{1}{4}, \frac{3}{4}, 1; \frac{1}{p^2}\right) \quad (17)$$

As in Appendix B, the generalized on-axis gradient* is defined to be

$$g_m(z - z') = m \lim_{r \rightarrow 0} \frac{U_m(r, z - z')}{r^m} \quad (18)$$

and noting that in the limit $r \rightarrow 0$ we have $1/p^2 = 0$ and $F = 1$, we get

$$g_m(z - z') = \lambda_m a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \frac{a^m}{[a^2 + (z-z')^2]^{m+1/2}} \quad (19)$$

*Usually, the factor of m is omitted from the definition, but is included here and in Marylie subroutines to make the result agree with the usual definition of the quadrupole gradient, for which $m = 2$. However, in order to define an on-axis gradient for the $m = 0$ case, the traditional definition without the factor of m must be used.

If we now consider an extended (but bounded or z -integrable) areal charge density of the form $\sigma_m(z') \sin m\phi$ on the cylinder, we get the on-axis gradient by integration over z' :

$$g_m(z) = a \frac{a^m (2m-1)!!}{2^{m+1} (m-1)!} \int_{-\infty}^{+\infty} \frac{\sigma_m(z')}{[a^2 + (z - z')^2]^{m+1/2}} dz' \quad (20)$$

If $\sigma_m(z')$ is represented by, for example, piecewise-continuous polynomials, it is straightforward to perform the integration in Eq. 20 analytically. Deriving analytical expressions for multiple derivatives of the right-hand side of Eq. 20 is also straightforward.

4 Gradient-length product, or z -integrated generalized on-axis gradient

In specifying the strength of magnets for input to beam-optics codes, it is convenient to use the idea of a gradient-length product, gL_{eff} . Going back to Eq. 19, we integrate the right-hand side from $-\infty$ to $+\infty$ in z . Using, for example, Maxima, we have

$$\int_{-\infty}^{+\infty} \frac{dz}{[a^2 + z^2]^{m+1/2}} = \frac{2^m (m-1)!}{(2m-1)!! a^{2m}} \quad (21)$$

and we have for the bounded extended charge distribution of Eq. 20

$$gL_{\text{eff}} \equiv \int_{-\infty}^{+\infty} g_m(z) dz = \frac{1}{2a^{m-1}} \int_{-\infty}^{+\infty} \sigma_m(z) dz \quad (22)$$

5 Connection between a radially-pointing surface dipole density, stream function, and surface currents

The concept of a stream function, first introduced by Lagrange in 1781, was an early development in the theory of two-dimensional incompressible flow. Since the surface of a cylinder is developable, the concept applies also to currents on the surface of a cylinder of radius a , where instead of the variables x and y , the appropriate variables are $a\phi$, z . Of course, a stream function on the surface of a cylinder must be periodic in ϕ . Here we show that the stream function on a cylinder can be interpreted as a surface density of radially-pointing dipoles (for more on this topic, see, e.g. Ref. [11]). This, in turn, allows construction of a scalar potential based on the field of an infinitesimal dipole.

By definition, a stream function $\psi(a\phi, z)$ produces surface currents

$$j_\phi = \frac{\partial \psi}{\partial z} \quad j_z = -\frac{1}{a} \frac{\partial \psi}{\partial \phi} \quad (23)$$

In order to establish the connection between dipole densities (i.e. magnetization) and currents, we start with the three-dimensional problem. In the case of volume magnetization $\mathbf{M}$, the equivalent volume current density is $\mathbf{J}_{\text{vol}} = \nabla \times \mathbf{M}$. We note that any current density derived from the curl of a vector function is automatically divergence-free. Now suppose $\mathbf{M}$ is of the form

$$\mathbf{M} = f(a, \phi, z) \hat{a} \quad (24)$$

where a is now variable. Then

$$\mathbf{J}_{\text{vol}} = \nabla \times \mathbf{M} = \frac{\partial f}{\partial z} \hat{\phi} - \frac{1}{a} \frac{\partial f}{\partial \phi} \hat{z} \quad (25)$$

We see that this $\mathbf{J}_{\text{vol}}$ has no terms containing derivatives with respect to a . Since $\mathbf{M}$ is periodic in ϕ , we can decompose it into Fourier components. Excluding axisymmetric terms, we can write

$$\mathbf{M} = \sum_{m=0}^{\infty} f_m(a, z) \sin m\phi \hat{a} + \text{cosine terms} \quad (26)$$

Now suppose $\mathbf{M}$ has only one Fourier component:

$$\mathbf{M} = f_m(a, z) \sin m\phi \hat{a} \quad (27)$$

Then

$$\mathbf{J}_{\text{vol}} = \nabla \times \mathbf{M} = \hat{a} \frac{\partial f_m}{\partial z} \sin m\phi \hat{\phi} - \frac{m}{a} f_m \cos m\phi \hat{z} \quad (28)$$

We now note (without proof at this point) that Eq. 28 can be used to construct radially thick windings that produce pure multipole fields in either open or axisymmetric boundaries. Ignoring the connections between turns, and modeling the windings as a set of closed turns, a single turn is defined with the prescription $a = \text{const.}$, $f_m(a, z) \sin m\phi = \text{const.}$ (see Ref. [12]).

We go from volume currents to surface currents by setting $\mathbf{M}_a = \psi(\phi, z) \delta(a - a') \hat{a}$, where δ is the Dirac delta function. Then the surface dipole density is given by integration of $\mathbf{M}_a$ in radius:

$$\mathbf{m} = \int \mathbf{M}_a da' = \psi(\phi, z) \hat{a} \quad (29)$$

The volume current from $\mathbf{M}_a$ is given by

$$\mathbf{J}_a = \nabla \times \mathbf{M}_a = \left[\frac{\partial \psi}{\partial z} \hat{\phi} - \frac{1}{a} \frac{\partial \psi}{\partial \phi} \hat{z} \right] \delta(a - a') \quad (30)$$

The surface current is then given by integration of $\mathbf{J}_a$ in radius:

$$\mathbf{j} = \int \mathbf{J}_a da' = \frac{\partial \psi}{\partial z} \hat{\phi} - \frac{1}{a} \frac{\partial \psi}{\partial \phi} \hat{z}, \quad (31)$$

which agrees with Eqs. 23. When ψ is used to represent real surface currents, it is convenient to add to the stream function for a pure multipole a factor of NI , which represents the number of ampere-turns per pole (see the following section).

6 Derivation of the scalar potential and on-axis gradient for a sinusoidally varying radially-pointing dipole density

In the language of classical potential theory, Sections 3 and 4 dealt with the single-layer open-boundary potential. We now turn to the double-layer potential, which in the cylindrical-surface case is the potential of a distribution of radially-pointing infinitesimal dipoles on the surface. Note that in this memo, the terms radially-pointing dipole layer, stream function, and double layer are used interchangeably. A radially pointing dipole distribution (double layer) can be used in two different ways in modeling magnetic fields:

1. To represent a real surface-current distribution in constructing approximate magnet models.
2. To represent a fictitious surface-current distribution in reconstructing magnetic fields from surface magnetic field data by use of the Helmholtz decomposition (see the later Section 9).

In the case of the fictitious currents of the Helmholtz decomposition, ψ has units of T-m (tesla-meters). In the case of real surface currents, ψ has units of amperes, and if the dipole distribution is written in the form $\psi(\phi, z)\hat{a}$, the surface currents are

$$j_\phi = \frac{\partial \psi}{\partial z} \quad j_z = -\frac{1}{a} \frac{\partial \psi}{\partial \phi} \quad (32)$$

That is, ψ is the stream function for surface currents in the two-dimensional developed cylindrical surface. Next, assuming there are no solenoidal currents, ψ can be expressed as a Fourier series in ϕ :

$$\psi(\phi, z) = \sum_{m=1}^{\infty} [f_m(z) \sin m\phi + g_m(z) \cos m\phi] \quad (33)$$

In accelerator-magnet parlance, the f_m terms represent "normal" multipoles and the g_m terms represent "skew" multipoles.* If ψ contains only one multipole term, the overall field produced by the windings will have pure $\sin m\theta$ or $\cos m\theta$ symmetry in either open boundaries or axisymmetric closed boundaries.

Usually, if $f_m(z)$ or $g_m(z)$ is intended to represent real currents, an explicit factor of NI , where NI is the number of ampere-turns per pole, is added to the stream function. Then for a pure normal multipole winding we can write

$$\psi(\phi, z) = NI f_m(z) \sin m\phi \quad (34)$$

With the NI factor in ψ , $f_m(z)$ is dimensionless and has a maximum value of 1. The peak j_z magnitude occurs at z values where $f_m(z) = 1$ and at $\phi = 0, \pi/m, 2\pi/m, \dots, 2\pi$ and is $j_z^{\max} = mNI/a$. At $\phi = 0$ j_z is negative and at π/m it is positive, etc. The maximum j_ϕ occurs at z values near or at the ends of the windings where $|df/dz|$ is maximum and at ϕ values $\phi = \pi/(2m), 3\pi/(2m), 5\pi/(2m), \dots$. The maximum j_ϕ at these points is $NI|df/dz|^{\max}$.

The continuous windings represented by the stream function can be discretized by setting $f_m(z) \sin m\phi = c_n, n = 1, 2, \dots, N$. Then individual turns are defined by such contours of constant ψ . Approximately, $c_n = (n - 1/2)/N$ (see Ref. 12).

In deriving expressions for potentials and gradients, we start with a ring of radius a with a line dipole density $\chi = \chi_m \sin m\phi$ at z' . Then the scalar potential is

$$V_m(r, \theta, z) = -\frac{\chi_m a}{4\pi} \int_0^{2\pi} \frac{\hat{a} \cdot (\mathbf{r} - \mathbf{r}') \sin m\phi d\phi}{|\mathbf{r} - \mathbf{r}'|^3} = -\frac{\chi_m a}{4\pi} \int_0^{2\pi} \frac{[r \cos(\phi - \theta) - a] \sin m\phi d\phi}{R^3}, \quad (35)$$

where again R is the distance between the field point r, θ, z and the source point a, ϕ, z' . The negative sign in Eq. 35 is chosen to ensure that with a positive dipole density at the center of the

*The overall sign in Eq. 32 is chosen to give a positive radial field at the first pole of a normal multipole winding when $f_m(z)$ is positive.

first pole, B_r for a normal multipole with be positive at the center of the pole. Setting $\alpha = \phi - \theta$ as before,

$$V_m^\sigma(r, \theta, z) = -\frac{\chi_m a}{2\pi} \sin m\theta \int_0^\pi d\alpha \frac{(r \cos \alpha - a) \cos m\alpha}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{3/2}} \quad (36)$$

We now note that Eq. 36 can be written in the form

$$V_m(r, \theta, z) = -\frac{\chi_m a}{2\pi} \sin m\theta \frac{\partial}{\partial a} \int_0^{2\pi} \frac{\cos m\alpha}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{1/2}} d\alpha \quad (37)$$

For an extended areal dipole density $f_m(z') \sin m\phi$, the scalar potential is given by a double integral:

$$V_m(r, \theta, z) = -\frac{a}{2\pi} \sin m\theta \int_0^\pi d\alpha \int_{-\infty}^{+\infty} \frac{(r \cos \alpha - a) \cos m\alpha f_m(z')}{[r^2 + a^2 - 2ar \cos \alpha + (z - z')^2]^{3/2}} dz' d\alpha \quad (38)$$

In general, it is impossible to evaluate the double integral of Eq. 38 in closed form; the comments on numerical evaluation of the double integrals for the potential and field components for the charge distribution of Section 2 also apply here.

Returning to a dipole ring at z' , the reduced scalar potential is

$$U_m(r, z) = -\frac{\chi_m a}{2\pi} \frac{\partial}{\partial a} \left[\frac{1}{\sqrt{ar}} Q_{m-1/2}(p) \right] = -\frac{\chi_m a}{2\pi} \frac{\partial}{\partial a} \left\{ \frac{1}{\sqrt{ar}} Q_{m-1/2} \left[\frac{a^2 + r^2 + (z - z')^2}{2ar} \right] \right\} \quad (39)$$

Taking the a derivative as indicated gives

$$U_m(r, z) = \frac{\chi_m a}{2\pi} \frac{1}{\sqrt{ar}} \left[\frac{1}{2a} Q_{m-1/2}(p) - \frac{d}{dp} Q_{m-1/2}(p) \frac{\partial p}{\partial a} \right] \quad (40)$$

From Ref. [3], p. 161, **3.8** (10), we have

$$\frac{d}{dp} Q_{m-1/2}(p) = \frac{m-1/2}{p^2-1} [p Q_{m-1/2}(p) - Q_{m-3/2}(p)] \quad (41)$$

and with

$$\frac{\partial p}{\partial a} = \frac{a^2 - r^2 - z^2}{2a^2 r} \quad (42)$$

we get the result

$$U_m(r, z) = \frac{\chi_m a}{2\pi} \frac{1}{\sqrt{ar}} \left\{ \frac{1}{2a} Q_{m-1/2}(p) - \frac{m-1/2}{p^2-1} [p Q_{m-1/2}(p) - Q_{m-3/2}(p)] \frac{a^2 - r^2 - z^2}{2a^2 r} \right\} \quad (43)$$

Figure 2 is a plot of $U_m(r, z)$ vs. r for $a = 1.0$, $z' = 0$, $m = 2$ and various values of z for a dipole ring with $\chi_m = 1$.

When $Q_{m-1/2}$ is expressed as a hypergeometric series, we get (see Eq. 16)

$$U_m(r, z) = -\chi_m a \frac{(2m-1)!!}{2^{m+1}m!} r^m \frac{\partial}{\partial a} \left\{ \frac{a^m}{[a^2 + r^2 + (z - z')^2]^{m+1/2}} F \left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; \frac{1}{p^2} \right) \right\} \quad (44)$$

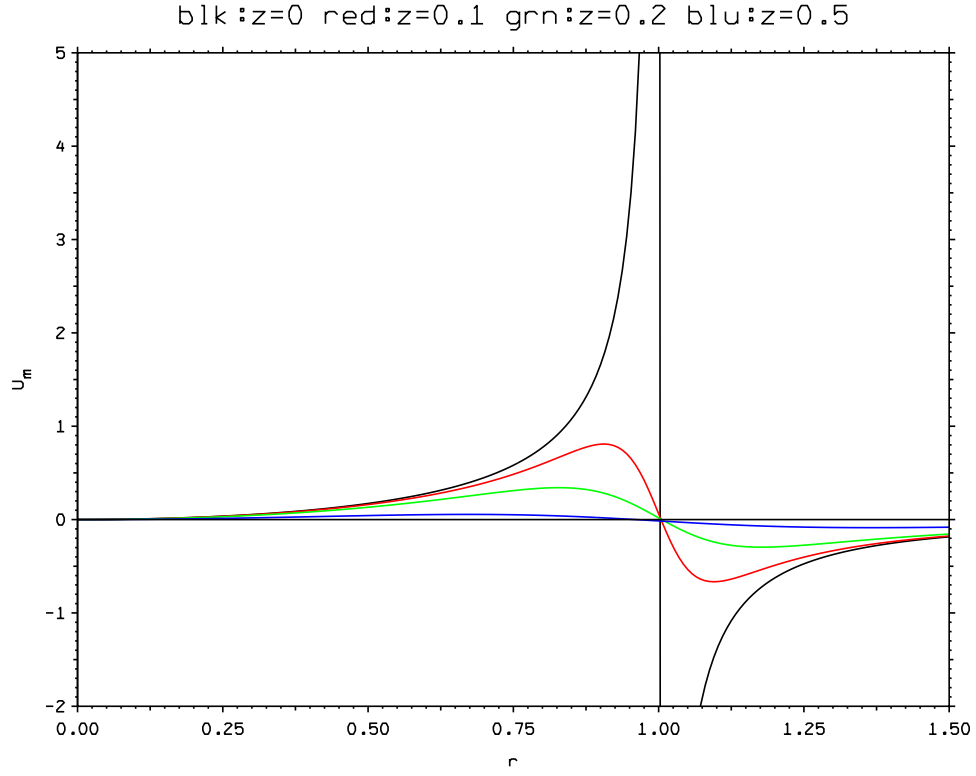


Figure 2: Plot of U_m vs. r for a radially-pointing dipole ring with $m = 2$ and $a = 1$ for various values of z . Black: $z = 0$; red: $z = 0.1$; green: $z = 0.2$; blue: $z = 0.5$.

To evaluate the a derivative, we will need $\partial(1/p^2)/\partial a$, which is given by

$$\frac{\partial}{\partial a} \left(\frac{1}{p^2} \right) = \frac{8ar^2(r^2 + z^2 - a^2)}{[a^2 + r^2 + (z - z')^2]^3} \quad (45)$$

We will also need the a derivative of the factor in front of the hypergeometric function:

$$\frac{\partial}{\partial a} \frac{a^m}{[a^2 + r^2 + (z - z')^2]^{m+1/2}} = a^{m-1} \frac{m[r^2 + (z - z')^2] - (m+1)a^2}{[a^2 + r^2 + (z - z')^2]^{m+3/2}} \quad (46)$$

Differentiation of the hypergeometric function with respect to $1/p^2$ just gives another hypergeometric function times a constant factor:

$$\frac{d}{d(1/p^2)} F\left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; \frac{1}{p^2}\right) = \frac{(\frac{m}{2} + \frac{1}{4})(\frac{m}{2} + \frac{3}{4})}{m+1} F\left(\frac{m}{2} + \frac{5}{4}, \frac{m}{2} + \frac{7}{4}, m+2; \frac{1}{p^2}\right) \quad (47)$$

Putting all of this together gives

$$U_m(r, z) = -\chi_m a \frac{(2m-1)!!}{2^{m+1}m!} r^m \left[a^{m-1} \frac{m[r^2 + (z - z')^2] - (m+1)a^2}{[a^2 + r^2 + (z - z')^2]^{m+3/2}} F\left(\frac{m}{2} + \frac{1}{4}, \frac{m}{2} + \frac{3}{4}, m+1; 1/p^2\right) + \frac{8a^{m+1}r^2[r^2 + (z - z')^2 - a^2]}{[a^2 + r^2 + (z - z')^2]^{m+7/2}} \frac{(\frac{m}{2} + \frac{1}{4})(\frac{m}{2} + \frac{3}{4})}{m+1} F\left(\frac{m}{2} + \frac{5}{4}, \frac{m}{2} + \frac{7}{4}, m+2; 1/p^2\right) \right] \quad (48)$$

Following the steps in the previous section for the gradient, we get for the generalized on-axis gradient for a ring with a sinusoidally varying radially-pointing dipole density

$$g_m(z - z') = m \lim_{r \rightarrow 0} \frac{U_m(r, z - z')}{r^m} = -\chi_m a \frac{(2m-1)!!}{2^{m+1}(m-1)!} a^{m-1} \frac{m(z - z')^2 - (m+1)a^2}{[a^2 + (z - z')^2]^{m+3/2}} = -\chi_m a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z - z')^2]^{m+1/2}} \quad (49)$$

As in the case of a surface charge distribution, if we now consider an extended areal dipole density of the form $f_m(z') \sin m\phi$ on the cylinder, we get the on-axis gradient by integration over z' :

$$g_m(z) = -a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} f_m(z') \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z - z')^2]^{m+1/2}} dz' \quad (50)$$

If $f_m(z')$ is represented by, for example, piecewise-continuous polynomials, it is straightforward to perform the integration in Eq. 50 analytically. We get an expression for the gradient-length product by following the steps used in derivation of the gradient-length product of a charge sheet (see Section 3). Using Eq. 21, we have

$$\int_{-\infty}^{+\infty} \frac{a^m dz}{[a^2 + z^2]^{m+1/2}} = \frac{2^m(m-1)!}{(2m-1)!!a^m} \quad (51)$$

Differentiating the right-hand side of Eq. 51 with respect to a , and applying the negative sign, we have

$$-\frac{\partial}{\partial a} \int_{-\infty}^{+\infty} \frac{a^m dz}{[a^2 + z^2]^{m+1/2}} = \frac{2^m m!}{(2m-1)!!a^{m+1}} \quad (52)$$

and from Eqs. 50 and 52 the gradient-length product for a dipole-sheet multipole is

$$gL_{\text{eff}} = \frac{m}{2a^m} \int_{-\infty}^{+\infty} f_m(z) dz \quad (53)$$

The function $f_m(z)$ in Eqs. 50 and 53 as written has units of T-m. If $f_m(z)$ is dimensionless as in Eq. 34 (i.e., $f(m)$ is the shape function for a current distribution) then we must add a factor of $\mu_0 NI$ to the expressions for the gradient and gradient-length product. Then Eq. 50 becomes

$$g_m(z) = -\mu_0 NI a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} f_m(z') \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z-z')^2]^{m+1/2}} dz' \quad (54)$$

and Eq. 53 becomes

$$gL_{\text{eff}} = \mu_0 NI \frac{m}{2a^m} \int_{-\infty}^{+\infty} f_m(z) dz \quad (55)$$

7 Derivation of the scalar potential and on-axis gradient for a ring with a dipole density with the Halbach rotation

We now consider a ring with a constant magnitude of linear magnetization density M_L (M_L has units of T-m² here) and a magnetization vector with the following angular dependence:

$$\mathbf{M}_L = -M_L \cos m\phi \hat{\phi} + M_L \sin m\phi \hat{a} \quad (56)$$

We note that the $\hat{a}$ term in Eq. 56 was already treated in Section 6. However, the $\hat{a}$ component in that section represented either surface currents or fictitious sources in the Helmholtz decomposition; here the $\hat{a}$ component, along with the $\hat{\phi}$ component, represents magnetization of real permanent-magnet material. We now find the contribution to the magnetic scalar potential of the combined $\hat{a}$ and $\hat{\phi}$ terms. We start with the expression

$$\begin{aligned} V_m(r, \theta, z) = & -\frac{M_L a}{4\pi} \int_0^{2\pi} \frac{(\sin m\phi \hat{a} - \cos m\phi \hat{\phi}) \cdot (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d\phi = \\ & -\frac{M_L a}{4\pi} \int_0^{2\pi} \frac{\{[r \cos(\phi - \theta) - a] \sin m\phi + r \sin(\phi - \theta) \cos m\phi\}}{[a^2 + r^2 + z^2 - 2ar \cos(\phi - \theta)]^3} d\phi \end{aligned} \quad (57)$$

As before, we set $\alpha = \phi - \theta$ and get the result

$$V_m(r, \theta, z) = -\frac{M_L a}{4\pi} \sin m\theta \int_0^{2\pi} \frac{(r \cos \alpha - a) \cos m\alpha - r \sin \alpha \sin m\alpha}{[a^2 + r^2 + (z - z')^2 - 2ar \cos \alpha]^{3/2}} d\alpha \quad (58)$$

We already evaluated the first term in Eq. 58 in Section 7. We use integration by parts to evaluate the second term:

$$\int_0^{2\pi} \frac{r \sin \alpha \sin m\alpha}{[a^2 + r^2 + (z - z')^2 - 2ar \cos \alpha]^{3/2}} d\alpha = \frac{m}{a} \int_0^{2\pi} \frac{\cos m\alpha}{[a^2 + r^2 + (z - z')^2 - 2ar \cos \alpha]^{1/2}} d\alpha \quad (59)$$

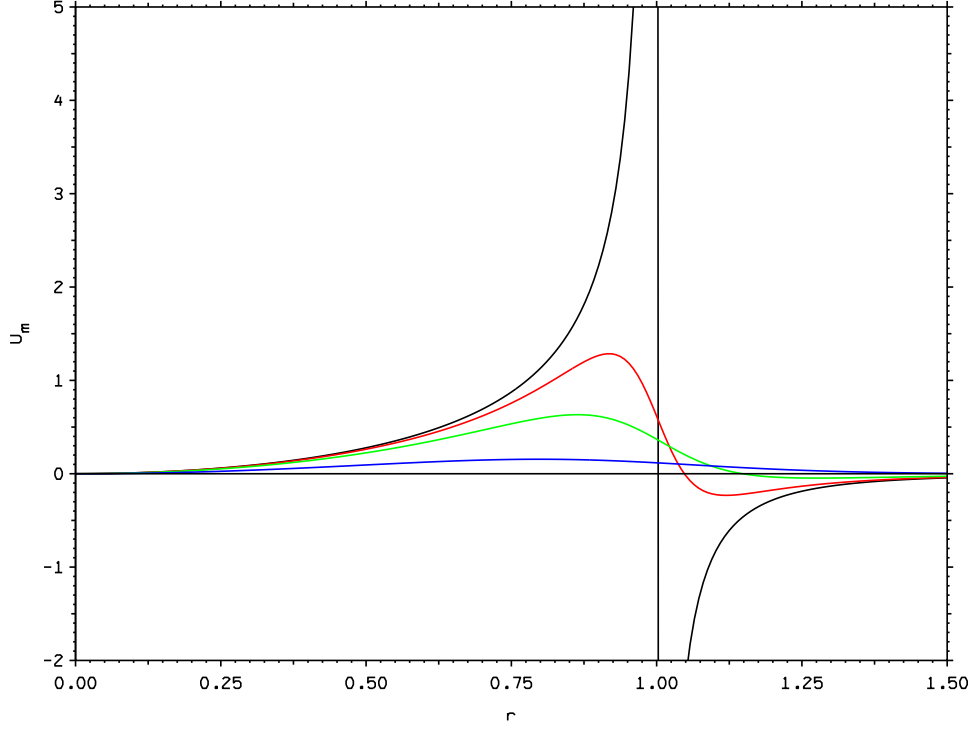


Figure 3: Plot of U_m vs. r for a Halbach ring with $m = 2$ and $a = 1$ for various values of z . Black: $z = 0$; red: $z = 0.1$; green: $z = 0.2$; blue: $z = 0.5$.

Since the integral from 0 to 2π is twice the integral from 0 to π , the total reduced potential can be written as

$$U_m(r, z) = \frac{M_L a}{2\pi} \left(\frac{m}{a} - \frac{\partial}{\partial a} \right) \int_0^\pi \frac{\cos m\alpha}{[a^2 + r^2 + (z - z')^2 - 2ar \cos \alpha]^{1/2}} d\alpha \quad (60)$$

and from Eqs. 12 and 13,

$$U_m(r, z) = \frac{M_L a}{2\pi} \left(\frac{m}{a} - \frac{\partial}{\partial a} \right) \frac{1}{\sqrt{ar}} Q_{m-1/2} \left[\frac{a^2 + r^2 + (z - z')^2}{2ar} \right] \quad (61)$$

Figure 3 is a plot of $U_m(r, z)$ vs. r for $a = 1.0$, $z' = 0$, $m = 2$ and various values of z for a Halbach ring with $M = 1$.

Following the steps in the previous section for the gradient, we get for the generalized on-axis gradient for a Halbach ring

$$\begin{aligned} g_m(z - z') &= m \lim_{r \rightarrow 0} \frac{U_m(r, z - z')}{r^m} = M_L a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \left(\frac{m}{a} - \frac{\partial}{\partial a} \right) \frac{a^m}{[a^2 + (z - z')^2]^{m+1/2}} \\ &= M_L a \frac{(2m+1)!!}{2^{m+1}(m-1)!} \frac{a^{m+1}}{[a^2 + (z - z')^2]^{m+3/2}} \end{aligned} \quad (62)$$

Using the integral of Eq. 21 except with $m+1$ in place of m , we get for the gradient-length product of the Halbach ring

$$gL_{\text{eff}} = \frac{mM_L}{a^m} \quad (63)$$

If we have a flat distribution of surface magnetization density M_S of length L , we get

$$gL_{\text{eff}} = \frac{mM_S}{a^m} L \quad (64)$$

Comparing gL_{eff} from Eq. 64 with gL_{eff} from Eq. 53, we see that the Halbach-ring gradient-length product has twice the gradient-length product of the radially-pointing dipole ring. This is the result of the fact that the Halbach ring is actually the sum of two rings: a radially-pointing dipole ring and a tangentially pointing (ϕ direction) ring that is rotated in angle by $\pi/(2m)$ (see Eq. 56). The two components contribute equally to gL_{eff} but not to $g_m(z)$ at a given point. The $g_m(z)$ components are equal in the center of long magnets where the field is essentially two-dimensional.

Finally, if we have a radially thick Halbach permanent magnet multipole with constant volume magnetization density M_V , length L , inner radius a_1 , and outer radius a_2 , the gradient-length product is

$$gL_{\text{eff}} = mM_V L \ln \frac{a_2}{a_1}, \quad m = 1$$

$$gL_{\text{eff}} = \frac{mM_V L}{m-1} \left(\frac{1}{a_1^{m-1}} - \frac{1}{a_2^{m-1}} \right), \quad m > 1 \quad (65)$$

If the magnitude of magnetization is constant, the expressions for the on-axis gradient and its z derivatives for the Halbach ring can be doubly integrated analytically over a rectangular cross section in r and z to get the on-axis gradient and its derivatives for a thick Halbach multipole magnet (see Ref. 15, Eqs. 7.14-7.18).

Incidentally, for the special case of a thick permanent-magnet quadrupole with the magnetization rotation of Eq. 56, in the limit of large a_2 , the gradient-length product becomes $gL_{\text{eff}} = 2M_V L/a_1$, and for large L , the gradient inside the bore away from the ends becomes $g = 2M_V/a_1$, i.e. with $M_V = B_{\text{rem}}$, the pole-tip field is $2B_{\text{rem}}$, where B_{rem} is the remanent field of the permanent-magnet material. This can be compared with a one-sided linear permanent magnet array with the rotation $\mathbf{M} = B_{\text{rem}}[-\sin kx\hat{x} + \cos kx\hat{y}]$, where the maximum field (which is at the upper surface of the array) is B_{rem} .

8 Expression for the vector potential from the scalar potential

If we have a vector potential $A_r\hat{r} + A_\theta\hat{\theta} + A_z\hat{z}$, the magnetic field is given by its curl:

$$B_r = \frac{1}{r} \frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z} \quad B_\theta = \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \quad B_z = \frac{1}{r} \frac{\partial(rA_\theta)}{\partial r} - \frac{1}{r} \frac{\partial A_r}{\partial \theta} \quad , \quad (66)$$

If we choose a gauge with $A_\theta = 0$, this simplifies to

$$B_r = \frac{1}{r} \frac{\partial A_z}{\partial \theta} \quad B_\theta = \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \quad B_z = -\frac{1}{r} \frac{\partial A_r}{\partial \theta} \quad (67)$$

We now assume that we have analytic solutions for the scalar potential in terms of a multipole expansion of the form of Eq. 1. If we equate B_r and B_z from Eqs. 2 and 67, we have

$$\frac{1}{r} \frac{\partial A_r}{\partial \theta} = -\frac{\partial U_m}{\partial z} \sin m\theta \quad \frac{1}{r} \frac{\partial A_z}{\partial \theta} = \frac{\partial U_m}{\partial r} \sin m\theta \quad (68)$$

Integration over θ gives

$$A_r = \frac{r}{m} \frac{\partial U_m}{\partial z} \cos m\theta \quad A_z = -\frac{r}{m} \frac{\partial U_m}{\partial r} \cos m\theta \quad (69)$$

The fact that the above gauge gives the correct B_θ is easily shown to be true using the fact that since $U_m \sin m\theta$ is a potential function by assumption, its Laplacian is zero.

9 Helmholtz theorem for a cylindrical surface

Approximate magnet models based on stream functions to represent currents, with and without boundaries, can be used to compute transfer maps by numerical integration, but some users will want to exactly model magnets, using either outputs of finite-element magnet-modeling computer programs, or measurements of magnetic fields of real magnets. The Helmholtz theorem provides a method of computing fields and on-axis gradients from fictitious sources derived from magnetic field data on a reference cylinder inside of the bore of the magnet. The important feature of this approach is that the interior fields are computed from the fictitious sources with open boundary conditions, regardless of the physical boundaries that came into play in producing the surface fields. The open boundaries in turn allow use of the simple integral kernels described earlier for evaluation of fields, potentials, and on-axis gradients and their derivatives.

In Ref. [10] the Helmholtz theorem was used to derive fictitious sources on an arbitrary surface surrounding a source-free volume that reproduce the field inside the volume. A Dirac monopole vector potential was used to obtain a vector potential for the fictitious magnetic charge sources representing the normal component of field at the surface. However, in the previous section, it was shown that if an analytic expression in cylindrical coordinates r, θ, z for scalar potential in terms of multipoles is known, a vector potential in the gauge with $A_\theta = 0$ can easily be derived. Here, the approach used is to derive scalar potentials for both the fictitious charge and current surface sources of the Helmholtz decomposition and then use the method of the previous section to derive the vector potentials for them. With this approach, direct use of the Dirac monopole vector potential is not needed when the surface is an infinite cylinder.

According to the Helmholtz theorem, the fictitious sources are of two types: a fictitious surface charge density $-\hat{n} \cdot \mathbf{B}$ and a fictitious surface current density $-\hat{n} \times \mathbf{B}$ (both with units of T). In cylindrical coordinates

$$-\hat{n} \cdot \mathbf{B} = -B_r \quad -\hat{n} \times \mathbf{B} = B_z \hat{\phi} - B_\phi \hat{z} \quad (70)$$

These sources are decomposed into their Fourier angular components as in Eqs. 4:

$$B_r = \sum_{m=1}^{\infty} b_r^m(z) \sin m\phi \quad -\hat{n} \times \mathbf{B} = b_z^m \sin m\phi \hat{\phi} - b_\phi^m \cos m\phi \hat{z} \quad (71)$$

Accordingly, for a particular normal field Fourier component, the fictitious charge distribution is

$$\sigma_m(z) \sin m\phi = b_r^m(z) \sin m\phi \quad (72)$$

It is straightforward to derive a scalar potential for it (see Section 2).

On the other hand, surface-current sources, real or fictitious, most naturally lead to expressions for a vector potential in open boundaries based on the integral

$$\mathbf{A} = \frac{1}{4\pi} \int_S \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS' \quad (73)$$

(this is a different gauge from that of the previous section). However, in this section we will not use Eq. 73 to compute a vector potential, but instead derive an expression for a scalar potential arising from the fictitious current sources. The scalar potential from the equivalent charges will be added to it. Then the method of the previous section will be used to construct a vector potential. To get the scalar potential produced by the fictitious currents, we will employ a scalar stream function $\psi(\phi, z)$ (or, equivalently, a double-layer distribution or radially-pointing surface dipole density) which is constructed from the fictitious surface currents (i.e, the field components tangential to the surface).

We recall (see Section 5) that by definition surface currents are derived from the stream function by the cross derivatives:

$$J_\phi = \frac{\partial \psi}{\partial z} \quad J_z = -\frac{1}{a} \frac{\partial \psi}{\partial \phi} \quad (74)$$

It is shown in Ref. [10] that the stream function (double-layer distribution) corresponding to the fictitious currents is just the scalar potential on the surface. That is,

$$\psi(\mathbf{r}) = \int \mathbf{B}(\mathbf{r}') \cdot d\mathbf{r}' \quad (75)$$

where $\mathbf{r}'$ is a point on the surface. The path stays on the surface and goes from a point on the surface with $\mathbf{B} = 0$ to the source point. For a particular normal multipole, assuming the field goes to zero at $z = \pm\infty$, we can use as the starting point of the path integral the point $(\phi, z) = (0, z)$, where z is any value, since $B_z(0, z) = 0$ for any z for a normal multipole field component. It is now convenient to use for ψ a path integral from $\phi = 0$ to ϕ , with z constant. This gives

$$\psi_m(\phi, z) = a \int_0^\phi B_\phi^m(\phi', z) d\phi' = \frac{a}{m} b_\phi^m(z) \sin m\phi, \quad (76)$$

where $b_\phi^m(z)$ is defined by Eq. 4 (with ϕ in place of θ to reflect the fact that ψ is a source function). We note that with this choice of ψ , the b_z^m data were not used. If b_z^m were to be used in constructing ψ , it would be necessary to numerically integrate b_z^m . With the ψ of Eq. 76, Eqs. 23 give

$$J_\phi = \frac{a}{m} \frac{\partial b_\phi^m}{\partial z} \sin m\phi \quad J_z = -b_\phi^m \cos m\phi \quad (77)$$

We see that J_z in Eqs. 77 agrees with J_z in Eqs. 70, but that J_ϕ is not explicitly the same. However, using the fact that $\nabla \times \mathbf{B} = 0$, we have

$$\frac{1}{a} \frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z} = 0 \quad (78)$$

and for a normal multipole

$$\frac{1}{a} \frac{\partial B_z}{\partial \phi} = \frac{m}{a} b_z^m \cos m\phi \quad \frac{\partial B_\phi}{\partial z} = \frac{\partial b_\phi^m}{\partial z} \cos m\phi \quad (79)$$

and

$$\frac{\partial b_\phi^m}{\partial z} = \frac{m}{a} b_z^m \quad (80)$$

With this result J_ϕ in Eqs. 77 becomes $b_z^m \sin m\psi$ and we see that the stream function ψ of Eq. 76 gives effective currents that agree with the currents of the Helmholtz decomposition.

10 Combined gradient-length product for Helmholtz charge and double-layer sources

We recall that for an extended charge distribution with open boundaries the gradient-length product is given by Eq. 22 and that for a dipole-layer (double-layer) distribution the gradient-length product is given by Eq. 53. Also, according Eq. 72, the m th component of the fictitious charge density is $b_r^m(z)$. The m th component of the double-layer density is given by Eq. 76, i.e. $\psi_m(\phi, z) = (a/m)b_\phi^m(z) \sin m\phi$. We also note that if $b_r^m(z)$ and $b_z^m(z)$ are bounded in z , i.e., approach zero for large $|z|$, then

$$\int_{-\infty}^{+\infty} b_r^m(z) dz = \int_{-\infty}^{+\infty} b_\phi^m(z) dz = Aa^{m-1} \quad (81)$$

where A is a constant that depends on the particular magnetic-field distribution. Equation 81 follows from the fact the z integral of a bounded 3-D scalar potential in cylindrical coordinates is a solution of the 2-D Laplace equation in cylindrical coordinates. Putting all of this together, the gradient-length product is

$$gL_{\text{eff}} = \frac{1}{2a^{m-1}} \left[\int_{-\infty}^{+\infty} b_r^m(z) dz + \int_{-\infty}^{+\infty} b_\phi^m(z) dz \right] = \frac{1}{a^{m-1}} \int_{-\infty}^{+\infty} b_r^m(z) dz = \frac{1}{a^{m-1}} \int_{-\infty}^{+\infty} b_\phi^m(z) dz \quad (82)$$

We see that the contributions of b_r^m and b_ϕ^m are equal. We also note that the equality of the b_r^m and b_ϕ^m z integrals can serve as a useful validity check on input field data for numerical gradient calculations using the Helmholtz decomposition.

11 Numerical computation of the on-axis gradient and its z derivatives from user field data on a reference cylinder using the Helmholtz decomposition

Here I assume that B_r and B_ϕ have been measured or computed on a sufficiently fine grid of ϕ, z points on a reference cylinder of radius a , and then Fourier-analyzed in ϕ . This results in data sets $b_r^m(z_i)$ and $b_\phi^m(z_i)$. Usually, the m values needed are only the fundamental m ($m = 1$ for a dipole, 2 for a quadrupole, etc.) and one or two "allowed" harmonics, i.e. $3m$ and perhaps $5m$. Then, according to the Helmholtz theorem, $b_r^m(z) \sin m\phi$ is the effective surface charge density and $(a/m)b_\phi^m(z) \sin m\phi$ the effective surface dipole density (see Eqs. 72 and 76). The data sets for the various m values associated with a particular magnet are treated as separate magnets, with the on-axis gradients to be superimposed when the coefficients of the expansion of the vector potential are computed.

Next, the $b_r^m(z_i)$ and $b_\phi^m(z_i)$ values are converted to piecewise-continuous cubics. The cubics on the interval $[z_i, z_{i+1}]$ are in the variable $t = z - z_i$ and are of the form $f(z) = A + Bt + Ct^2 + Dt^3$. Depending upon the quality and noisiness of the data, the cubic coefficients could be obtained by an interpolating spline or a least-squares spline routine. To standardize the input data, the user will compute the polynomial coefficients and prepare files for input to Marylie. The files will consist of a header line containing the multipole number m and the reference radius a , followed by lines with 9 numbers: z_i , the 4 cubic coefficients A_r, B_r, C_r, D_r for b_r^m , and the 4 cubic coefficients $A_\phi, B_\phi, C_\phi, D_\phi$ for b_ϕ^m .

The gradient calculation for the Helmholtz sources is done by a subroutine called **helmgrad** that sums the contributions of the charge-sheet and dipole-sheet components over the intervals. The algorithms for computing the charge-sheet and dipole-sheet gradients and their z derivatives for an interval are described in Appendices C and D. Computing the charge-sheet contribution of an interval to the gradient requires a single call to a subroutine **grad** that computes for the i th interval, with $\Delta = z_{i+1} - z_i$,

$$I_m = \int_0^\Delta \frac{\alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4}{[a^2 + (z - z_i - t)^2]^{m+1/2}} dt \quad (83)$$

g and its repeated z derivatives up to some maximum number (typically 20). For the equivalent charge sheet, $\alpha_0 = A_r$, $\alpha_1 = B_r$, etc. Since cubics are used, α_4 is set to zero in calls to **grad**. Computation of the dipole-sheet contribution requires two calls to **grad**, one with an exponent of $m + 1/2$ as in Eq. 83 and another with an exponent of $m + 3/2$, as in Eq. D5 of Appendix D. Magnetic-field test data for the Helmholtz gradient routines were generated by a computer program that computes the magnetic field from a cylindrical filament Lambertson-type quadrupole coil (see Fig. 4). The coil was built up out of straight filaments and circular arcs without turn-to-turn connections. The coil was designed to have an (almost) pure normal quadrupole z -integrated field, but the three-dimensional field had strong "allowed" harmonics: $m = 6, m = 10$, etc. as a function of z . These harmonics were in fact considerably larger than those of a typical quadrupole magnet and therefore served as a good test of the Helmholtz algorithms when applied to the higher harmonics. The winding cylinder had a radius of 0.1 m; the data cylinder for the Helmholtz decomposition had a radius r_{ref} of 0.08 m. Interpolating piecewise-continuous cubics for $b_r^m(z_i)$ and $b_\phi^m(z_i)$ were derived from the output of the spline-interpolation routine **spine** (see Ref. [14], p. 86, **3.3**) and from the data sets themselves. The computed gradient and its z derivatives were then used with the off-axis expansion of Appendix B, Eqs. B10 to reconstruct fields (see Fig. 5).

12 Modified-Bessel-function Solutions for Scalar and Vector Potentials and Magnetic Field Components in Cylindrical Coordinates

Modified-Bessel-function solutions are easily applied to sources extended in z , and provide solutions for closed boundaries, i.e., cases where the windings are surrounded by a coaxial cylinder of infinitely permeable material. Explicit use of Bessel-function expressions for computation of magnetic potentials and fields at many arbitrary evaluation points involves repeated numerical evaluation of Fourier integrals and is not as computationally efficient as some other methods. However, after Bessel-function expressions for the magnetic scalar potential have been found for various

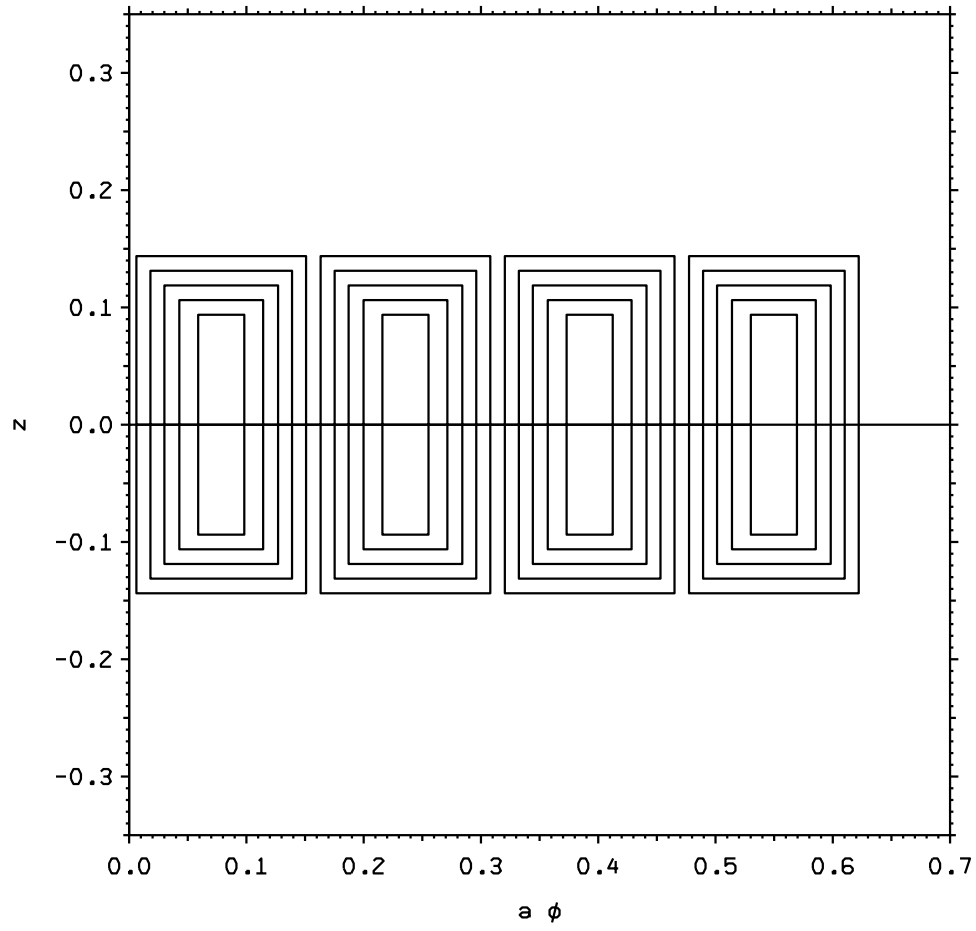


Figure 4: Developed Lambertson filament quadrupole winding

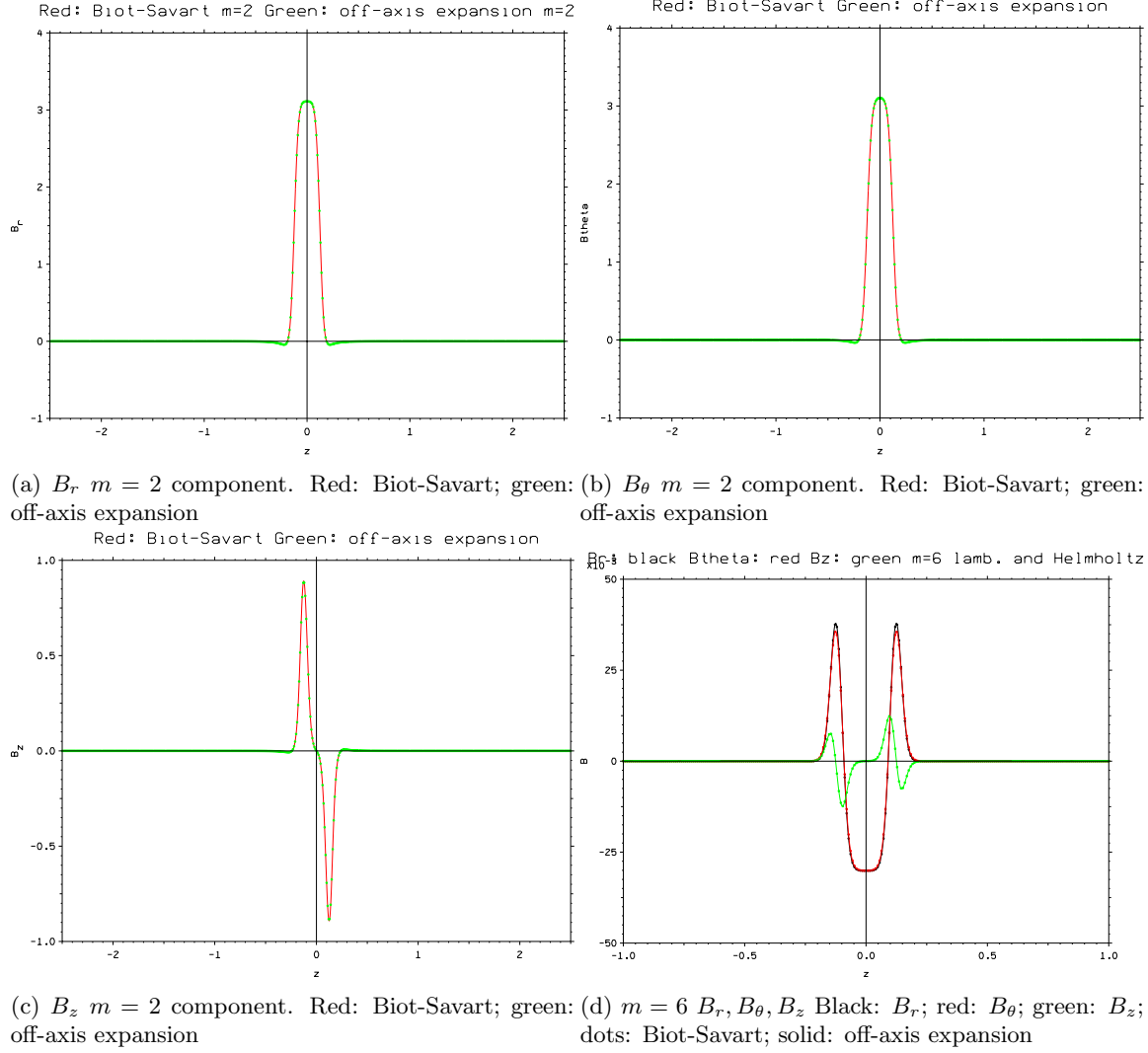


Figure 5: Comparison of Fourier components of the field from a Lambertson quadrupole computed directly from the Biot-Savart expression and reconstructed from the on-axis gradient and its derivatives by the off-axis expansion of Appendix B. The Lambertson coil radius was 0.1 m, the Helmholtz source cylinder radius 0.08 m, and the field-point radius 0.05 m.

source distributions, it is straightforward to find expressions for the field components and the vector potential. Moreover, computation time can be reduced by precomputing and storing the Fourier transforms of the source distributions. Algorithms based on Bessel-function Fourier integrals are therefore very useful in checking other numerical magnetic-field computation algorithms. Bessel-function integrals can also be used to find effective sources that, with open-boundaries, replicate the contribution of a closed boundary to the fields and gradient around the axis.

Separation of the Laplace equation in cylindrical coordinates leads to elementary modified-Bessel-function solutions of the form $R_m(kr)Z(kz)\Theta_m(\theta)$ with

$$\begin{aligned} R_m(kr) &= AI_m(kr) + BK_m(kr) \\ Z(kz) &= C \cos kz + D \sin kz \\ \Theta_m(\theta) &= E \sin m\theta + F \cos m\theta \end{aligned} \quad (84)$$

Alternatively, elementary solutions with ordinary Bessel functions, of the form

$$\begin{aligned} R_m(kr) &= AJ_m(kr) + BY_m(kr) \\ Z(kz) &= Ce^{kz} + De^{-kz} \\ \Theta_m(\theta) &= E \sin m\theta + F \cos m\theta \end{aligned} \quad (85)$$

are also valid and are useful in finding solutions for large $|z|$ with closed cylindrical boundaries. However, the rest of the section deals with the modified Bessel-function solutions. We note that in the limit $r \rightarrow 0$, $K_m(kr)$ approaches infinity, so near the z axis, the solution can contain only $I_m(kr)$. On the other hand, as $r \rightarrow \infty$, $I_m(kr)$ becomes infinite and K_m approaches zero, so with open boundaries, the solution can contain only $K_m(kr)$ for large r .

In the next sections, we apply boundary conditions to construct solutions for charge sheets and double layers based on the elementary solutions of Eq. 84.

12.1 Distribution of magnetic charges on a cylindrical surface, open boundaries

Let us consider a distribution of magnetic charges $\sigma(\phi, z)$ on a cylinder of radius a . Since real magnetic monopoles do not come into play in magnet modeling, the surface charge density considered here is fictitious and must be derived from the Helmholtz decomposition; σ has units of T. Next, we express σ as Fourier series in angle:

$$\sigma(\theta, z) = \sum_{m=1}^{\infty} [\sigma_m^s(z) \sin m\phi + \sigma_m^c(z) \cos m\phi] \quad (86)$$

Next, for brevity, we drop the cosine (skew multipole) terms; it is straightforward to put the skew terms back into the following development. We can now express σ_m as a Fourier integral in z . Defining

$$S_m^{\text{odd}}(k) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \sigma_m(z) \sin kz dz \quad S_m^{\text{even}}(k) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \sigma_m(z) \cos kz dz, \quad (87)$$

we can write

$$\sigma_m(z) = \int_0^{\infty} S_m^{\text{odd}}(k) \sin kz dk + \int_0^{\infty} S_m^{\text{even}}(k) \cos kz dk \quad (88)$$

In most magnet-modeling cases, we are dealing with magnetic distributions that are symmetric in z around the center of the magnet, and S_m^{odd} is zero. Therefore, for simplicity, we treat only the symmetric case and set $S_m = S_m^{\text{even}}$; the development for the antisymmetric case with $\sin kz$ in place of $\cos kz$ is straightforward.

We now consider the boundary conditions at a charge sheet (here a fictitious monopole sheet). From textbooks on elementary electromagnetism, we know that the scalar potential is continuous at a charge sheet, while the normal component of the field jumps by the quantity σ (with our units for σ). That is, in terms of the reduced potential,

$$U_m(a^+, z) - U_m(a^-, z) = 0 \quad (89)$$

$$-\frac{\partial U_m}{\partial r}(a^+, z) + \frac{\partial U_m}{\partial r}(a^-, z) = \sigma_m(z), \quad (90)$$

where a^+ and a^- refer to points just outside and just inside the cylinder, respectively. Next, we write down the general solution (for the symmetric-in- z case) in open coordinates of the reduced Laplace equation for $r < a$ and $r > a$ in the form of Fourier integrals, noting that the solution must approach zero at $r = a$ and $r = \infty$:

$$\begin{aligned} U_m(r, z) &= \int_0^\infty I_m(kr) K_m(ka) F_m^-(k) \cos kz dk, \quad r < a \\ U_m(r, z) &= \int_0^\infty I_m(ka) K_m(kr) F_m^+(k) \cos kz dk, \quad r > a \end{aligned} \quad (91)$$

Clearly, Eq. 89 is satisfied when $F_m^+ = F_m^- = F_m$. In order to satisfy Eq. 90, we must have

$$\int_0^\infty k [I_m'(ka) K_m(ka) - I_m(ka) K_m'(ka)] F(k) \cos kz dk = \int_0^\infty S_m(k) \cos kz dk \quad (92)$$

Finally, we use the Bessel-function Wronskian relation $I_m'(ka) K_m(ka) - I_m(ka) K_m'(ka) = 1/(ka)$ to get $F_m(k) = a S_m(k)$. Then the expression for the scalar potential is

$$U_m(r, z) = a \int_0^\infty I_m(kr) K_m(ka) S_m(k) \cos kz dk, \quad r < a \quad (93)$$

If the surface charge density is a delta function in z , $S_m(k) = 1/\pi$. Then, from tables of integrals (see Ref. [5], **6.672** 4., p. 732, Ref. [7, p. 57, Eq. 14b, or Ref. [8], 1.12 (47), p. 49), we find that

$$\int_0^\infty I_m(kr) K_m(ka) \cos kz dk = \frac{1}{2\sqrt{ar}} Q_{m-1/2} \left[\frac{a^2 + r^2 + z^2}{2ar} \right] \quad (94)$$

and replacing z with $z - z'$ to allow for a charge ring at $z = z'$ instead of $z = 0$, Eq. 112 becomes Eq. 13 with $\lambda_m = 1$.

The field components for the charge-sheet multipole magnet with open boundaries are given by the gradient in cylindrical coordinates times the appropriate $\sin m\theta$ or $\cos m\theta$ factor:

$$\begin{aligned} B_r(r, \theta, z) &= a \sin m\theta \int_0^\infty k I_m'(kr) K_m(ka) S_m(k) \cos kz dk, \quad r < a \\ B_\theta(r, \theta, z) &= a \cos m\theta \int_0^\infty mk \frac{I_m(kr)}{kr} K_m(ka) S_m(k) \cos kz dk, \quad r < a \\ B_z(r, \theta, z) &= -a \sin m\theta \int_0^\infty k I_m(kr) K_m(ka) S_m(k) \sin kz dk, \quad r < a \end{aligned} \quad (95)$$

We note that the ratio of the Bessel-function kernels for U_m and B_θ is independent of k and is r/m . Using Eqs. 69, we get for the vector potential

$$\begin{aligned} A_r(r, \theta, z) &= -a \cos m\theta \frac{r}{m} \int_0^\infty k I_m(kr) K_m(ka) S_m(k) \sin kz dk, \quad r < a \\ A_r(r, \theta, z) &= -a \cos m\theta \frac{r}{m} \int_0^\infty k I'_m(kr) K_m(ka) S_m(k) \cos kz dk, \quad r < a \end{aligned} \quad (96)$$

Using the definition for the on-axis generalized gradient of Eq. B12, we get for $g_m(z)$ for the charge-sheet multipole magnet by using the small- r limit for $I_m(kr)$ (see Eqs. 102 below) and taking the limit of $mU_m(r, z)/r^m$ as r approaches zero as in Eq. 112:

$$g_m(z) = a \frac{1}{2^m(m-1)!} \int_0^\infty k^m K_m(ka) S_m(k) \cos kz dk \quad (97)$$

Again, if $\sigma(z)$ is a delta function, $S_m(k) = 1/\pi$, and Eq. 97 becomes

$$g_m(z) = a \frac{1}{2^m(m-1)! \pi} \int_0^\infty k^m K_m(ka) \cos kz dk \quad (98)$$

The integral in Eq. 98 can be found in Ref. [5], **6.71** 12., p. 749. and in Ref. [8] (41), p. 49. The expression for integer m is

$$\int_0^\infty k^m K_m(ka) \cos kz dk = \frac{\pi}{2} a^m (2m-1)!! \frac{1}{(a^2 + z^2)^{m+1/2}} \quad (99)$$

Multiplying out the factors in Eqs. 98 and 99 gives

$$g_m(z) = a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \frac{a^m}{[a^2 + z^2]^{m+1/2}} \quad (100)$$

which is the same as Eq. 19 except for the factor λ_m and the shift in z to from z to $z - z'$. The z derivatives of g_m are obtained by differentiation inside the integral sign:

$$\begin{aligned} \frac{d^n}{dz^n} g_m(z) &= (-1)^{\frac{n+1}{2}} a \frac{1}{2^m(m-1)!} \int_0^\infty k^{m+n} K_m(ka) S_m(k) \sin kz dk, \quad n \text{ odd} \\ \frac{d^n}{dz^n} g_m(z) &= (-1)^{\frac{n}{2}} a \frac{1}{2^m(m-1)!} \int_0^\infty k^{m+n} K_m(ka) S_m(k) \cos kz dk, \quad n \text{ even} \end{aligned} \quad (101)$$

In numerically evaluating the integrals in Eqs. 112 and 95, we have to evaluate the factors $I_m(kr)K_m(ka)$, $kI'_m(kr)K_m(ka)$, $kI_m(kr)K_m(ka)/(kr)$, and $kI_m(kr)K_m(ka)$ for zero or small k , where K_m approaches infinity, but the factors are finite. In the following section concerning the fields from dipole sheets, we will also have to evaluate expressions involving $K'_m(ka)$. To do this we use the small-argument limits of $I_m(kr)$, $K_m(ka)$, $I'_m(kr)$, and $K'_m(ka)$ (see Ref. [1], **9.6.7** and **9.6.9**, p. 375) :

$$\begin{aligned} I_m(kr) &\sim \frac{1}{m!} \left(\frac{kr}{2} \right)^m & K_m(ka) &= \frac{(m-1)!}{2} \left(\frac{2}{ka} \right)^m \\ I'_m(kr) &\sim \frac{1}{2} \frac{1}{(m-1)!} \left(\frac{kr}{2} \right)^{m-1} & K'_m(ka) &\sim -2^{m-1} m! \frac{1}{(ka)^{m+1}} \end{aligned} \quad (102)$$

Then

$$I_m(kr)K_m(ka) \sim \frac{1}{2m} \left(\frac{r}{a}\right)^m \quad kI'_m(kr)K_m(ka) \sim \frac{1}{2a} \left(\frac{r}{a}\right)^{m-1}, \text{ small } k \quad (103)$$

$$mk \frac{I_m(kr)}{kr} K_m(ka) \sim \frac{1}{2a} \left(\frac{r}{a}\right)^{m-1}, \text{ small } k \quad (104)$$

We may want an expression for $\partial B_r / \partial r$. For this, we need

$$k^2 I''_m(kr)K_m(ka) \sim \frac{m-1}{2a^2} \left(\frac{r}{a}\right)^{m-2}, \text{ small } k \quad (105)$$

12.2 Gradient-length product for a distribution of magnetic charges on a cylinder, open boundaries

We can get an expression for the integrated gradient gL_{eff} by first integrating the right-hand side of Eq. 97 from $-\infty$ to $+\infty$ with respect to z . The integral over z of $\cos kz$ gives $2\pi\delta(k)$, i.e., 2π times a delta function centered at $k = 0$. Since the k integral goes from 0 to ∞ , only one half of the delta function contributes to the k integral. We then have

$$gL_{\text{eff}} = a \frac{1}{2^m(m-1)!} \pi \lim_{k \rightarrow 0} [k^m K_m(ka) S_m(k)] \quad (106)$$

From Eqs. 102 we have

$$\lim_{k \rightarrow 0} [k^m K_m(ka)] = \frac{2^{m-1}(m-1)!}{a^m} \quad (107)$$

and from Eq. 87

$$S_m(0) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \sigma_m(z) dz \quad (108)$$

Multiplying the factors together gives

$$gL_{\text{eff}} = \frac{1}{2a^{m-1}} \int_{-\infty}^{+\infty} \sigma_m(z) dz \quad (109)$$

which agrees with Eq. 22.

12.3 Distribution of magnetic charges on a cylindrical surface, closed cylindrical boundary (iron boundary)

Let us now consider a distribution of magnetic charges on a cylinder of radius a with a cylindrical boundary of infinitely permeable material at radius b . Since real magnetic monopoles apparently do not exist, this problem is hypothetical. However, with a change of units, the results of this section could be applied to electrostatic problems with real electrical charges. In that case, the closed boundary is an electrical conductor. As before, we express $\sigma(\phi, z)$ as Fourier series in angle and deal with a single component $\sigma_m(z)$ and its Fourier cosine transform $S_m(z)$.

We have the same boundary conditions at the charge sheet as in the previous section (see Eqs. 89 and 90). However, there is an additional condition at $r = b$, namely $U_m(b, z) = 0$. As before, the

solution can contain only $I_m(kr)$ for $r < a$ in order to satisfy the condition $U_m(0, z) = 0$. However, in order to satisfy the boundary conditions at both $r = a$ and $r = b$, the solution for $b > r > a$ must be a linear combination of $I_m(kr)$ and $K_m(kr)$. Therefore, the solution must be of the form

$$\begin{aligned} U_m(r, z) &= \int_0^\infty I_m(kr) A(k) \cos kz dk, & r < a \\ U_m(r, z) &= \int_0^\infty [B(k) I_m(kr) + C(k) K_m(kr)] \cos kz dk, & b > r > a \end{aligned} \quad (110)$$

The three functions $A(k)$, $B(k)$ and $C(k)$ are found by using the boundary conditions of Eqs. 89 and 90 together with the condition $U_m(r, b) = 0$. Again, we make use of the Bessel-function Wronskian relation $I'_m(ka)K_m(ka) - I_m(ka)K'_m(ka) = 1/(ka)$. The solution for $A(k)$ is

$$A(k) = a \left[K_m(ka) - \frac{I_m(ka)K_m(kb)}{I_m(kb)} \right] S_m(k) \quad (111)$$

and $U_m(r, z)$ for $r < a$ is

$$U_m(r, z) = a \int_0^\infty I_m(kr) \left[K_m(ka) - \frac{I_m(ka)K_m(kb)}{I_m(kb)} \right] S_m(k) \cos kz dk, \quad r < a \quad (112)$$

As a check, we note that in the limit $b \rightarrow \infty$, the second term in the brackets goes to zero, and Eq. 110 becomes the same as Eq. 112.

The expression for the on-axis gradient is derived in the same way as in the previous section but with $K_m(ka) - I_m(ka)K_m(kb)/I_m(kb)$ in place of $K_m(ka)$. The resulting expression is

$$g_m(z) = a \frac{1}{2^m(m-1)!} \int_0^\infty k^m \left[K_m(ka) - \frac{I_m(ka)K_m(kb)}{I_m(kb)} \right] S_m(k) \cos kz dk \quad (113)$$

There apparently is no closed-form expression for the integral in Eq. 113. The gradient-length product is obtained by following the steps used to find the gradient-length product in Section 12.2, but with the additional term in the kernel that takes into account the closed boundary:

$$gL_{\text{eff}} = a \frac{1}{2^m(m-1)!} \pi \lim_{k \rightarrow 0} \left\{ k^m \left[K_m(ka) - \frac{I_m(ka)K_m(kb)}{I_m(kb)} \right] S_m(k) \right\} \quad (114)$$

The result is

$$gL_{\text{eff}} = \frac{1}{2a^{m-1}} \left[1 - \left(\frac{a}{b} \right)^{2m} \right] \int_{-\infty}^{+\infty} \sigma_m(z) dz \quad (115)$$

We see that adding the boundary actually reduces the gradient-length product, which approaches zero as a approaches b . We will see in a later section that this is different from the effect of the boundary on the field for dipole sheets (current sheets), for which the infinitely permeable boundary increases the z -integrated fields and gradients.

12.4 Distribution of radially-pointing point magnetic dipoles on a cylindrical surface, open boundaries

In the language of mathematical potential theory, the radially-pointing surface dipole distribution is called a double layer. The boundary condition at a double layer is that the normal component

of field is continuous and the scalar potential jumps by a constant times the double-layer density ψ . In applying the Helmholtz decomposition (see Eq. 76), ψ has units of T-m. In the case of real currents on a cylinder, ψ has units of A (amperes) and the jump in the magnetic scalar potential is $\mu_0\psi$. In the case of real currents, ψ can be made dimensionless by taking out an additional factor of NI , where NI is the number of ampere-turns in a pole. Here we will not use a factor of μ_0NI in the expressions for fields and potentials and just note that μ_0NI must be included when ψ models real currents and is dimensionless.

For a particular multipole component, the jump in the reduced potential is

$$U_m(a^+, z) - U_m(a^-, z) = f_m(z) \quad (116)$$

where f_m is the m th component of the double-layer density. The condition that the normal component of field is continuous is

$$\frac{\partial U_m}{\partial r}(a^+, z) - \frac{\partial U_m}{\partial r}(a^-, z) = 0 \quad (117)$$

Let us again write the Bessel-function solutions for $r < a$ and $r > a$ as Fourier cosine integrals (again assuming a distribution symmetric in z), but without the factor of $K_m(ka)$ for $r < a$ and $I_m(ka)$ for $r > a$:

$$\begin{aligned} U_m(r, z) &= \int_0^\infty I_m(kr) F_m^-(k) \cos kz dk, \quad r < a \\ U_m(r, z) &= \int_0^\infty K_m(kr) F_m^+(k) \cos kz dk, \quad r > a \end{aligned} \quad (118)$$

The condition that the normal (radial) component of field is continuous implies that $kK'_m(ka)F_m^+(k) = kI'_m(ka)F_m^-(k)$. Next, we write $f_m(z)$ as as a Fourier integral:

$$C_m(k) = \frac{1}{\pi} \int_{-\infty}^{+\infty} f_m(z) \cos kz dz \quad f_m(z) = \int_0^\infty C_m(k) \cos kz dk \quad (119)$$

Then we have

$$\begin{aligned} U_m(a^+, z) - U_m(a^-, z) &= \int_0^\infty \frac{1}{K'_m(ka)} [I'_m(ka)K_m(ka) - K'_m(ka)I_m(ka)] F_m^-(k) \cos kz dk \\ &= \int_0^\infty C(k) \cos kz dk \end{aligned} \quad (120)$$

Again using the Bessel-function relation $I'_m(ka)K_m(ka) - I_m(ka)K'_m(ka) = 1/(ka)$, we have the result

$$\begin{aligned} U_m(r, z) &= -a \int_0^\infty k I_m(kr) K'_m(ka) C_m(k) \cos kz dk, \quad r < a \\ U_m(r, z) &= -a \int_0^\infty k I'_m(ka) K_m(kr) C_m(k) \cos kz dk, \quad r > a \end{aligned} \quad (121)$$

The function $C_m(k)$ has units of T-m² as it appears in Eqs. 121 and elsewhere in this section.

The field components for the current-sheet multipole magnet with open boundaries are given by the gradient in cylindrical coordinates times the appropriate $\sin m\theta$ or $\cos m\theta$ factor:

$$\begin{aligned} B_r(r, \theta, z) &= -a \sin m\theta \int_0^\infty k^2 I'_m(kr) K'_m(ka) C_m(k) \cos kz dk, \quad r < a \\ B_\theta(r, \theta, z) &= -a \cos m\theta \int_0^\infty mk^2 \frac{I_m(kr)}{kr} K'_m(ka) C_m(k) \cos kz dk, \quad r < a \\ B_z(r, \theta, z) &= a \sin m\theta \int_0^\infty k^2 I_m(kr) K'_m(ka) C_m(k) \sin kz dk, \quad r < a \end{aligned} \quad (122)$$

We again note that the ratio of the Bessel-function kernels for U_m and B_θ is independent of k and is r/m . Following the steps of the previous section, we get for $g_m(z)$ for the current-sheet multipole magnet

$$g_m(z) = -a \frac{1}{2^m(m-1)!} \int_0^\infty k^{m+1} K'_m(ka) C_m(k) \cos kz dk \quad (123)$$

The z derivatives of g_m are again obtained by differentiation inside the integral sign:

$$\begin{aligned} \frac{d^n}{dz^n} g_m(z) &= (-1)^{\frac{n-1}{2}} a \frac{1}{2^m(m-1)!} \int_0^\infty k^{m+n+1} K'_m(ka) C_m(k) \sin kz dk, \quad n \text{ odd} \\ \frac{d^n}{dz^n} g_m(z) &= (-1)^{\frac{n-2}{2}} a \frac{1}{2^m(m-1)!} \int_0^\infty k^{m+n+1} K'_m(ka) C_m(k) \cos kz dk, \quad n \text{ even} \end{aligned} \quad (124)$$

In numerically evaluating the integrals in Eqs. 121-122, we have to evaluate the factors $kI_m(kr)K'_m(ka)$, $k^2I'_m(kr)K'_m(ka)$, and $k^2I_m(kr)K'_m(ka)/(kr)$ for zero or small k , where K'_m approaches infinity, but the factors are finite. To do this we use the small-argument limits of $I_m(kr)$, $K_m(ka)$, $I'_m(kr)$, and $K'_m(ka)$ given in the previous section. Then

$$kI_m(kr)K'_m(ka) \sim -\frac{1}{2a} \left(\frac{r}{a}\right)^m \quad k^2I'_m(kr)K'_m(ka) \sim -\frac{m}{2a^2} \left(\frac{r}{a}\right)^{m-1}, \quad \text{small } k \quad (125)$$

$$mk^2 \frac{I_m(kr)}{kr} K'_m(ka) \sim -\frac{m}{2a^2} \left(\frac{r}{a}\right)^{m-1}, \quad \text{small } k \quad (126)$$

Also, we may want an expression for $\partial B_r / \partial r$. For this, we need

$$k^3 I''_m(kr) K'_m(ka) \sim -\frac{m(m-1)}{2a^3} \left(\frac{r}{a}\right)^{m-2}, \quad \text{small } k \quad (127)$$

12.5 Gradient-length product for a distribution of radially-pointing magnetic dipoles on a cylinder, open boundaries

To get the gradient-length product, we use the same limiting process as in Eq. 106, but with the integral kernel of Eq. 121 for $r < a$:

$$gL_{\text{eff}} = -a \frac{1}{2^m(m-1)!} \pi \lim_{k \rightarrow 0} [k^{m+1} K'_m(ka) C_m(k)] \quad (128)$$

From Eqs. 102 we have

$$\lim_{k \rightarrow 0} [k^{m+1} K'_m(ka)] = -\frac{2^{m-1}m!}{a^{m+1}} \quad (129)$$

and from Eq. 119

$$C_m(0) = \frac{1}{\pi} \int_{-\infty}^{+\infty} f_m(z) dz \quad (130)$$

The result for the gradient-length product of a distribution of radially-pointing magnetic dipoles on a cylinder with open boundaries is

$$gL_{\text{eff}} = \frac{m}{2a^m} \int_{-\infty}^{+\infty} f_m(z) dz \quad (131)$$

which agrees with Eq. 53.

12.6 Distribution of radially-pointing point magnetic dipoles on a cylindrical surface, closed cylindrical boundary (iron boundary)

In applying the Helmholtz theorem to field modeling, it was shown previously that a radially-pointing dipole layer (double layer) can be used to represent a stream function for surface currents on a cylinder. Solutions for the scalar potential and fields with open boundaries were derived both in terms of integrals in z over the source function and as Fourier integrals containing Bessel functions. In this section, Bessel-function solutions are derived for cylindrical current sheets with a surrounding coaxial boundary of infinite permeability and radius b (the infinitely permeable boundary will be referred to henceforth as the "iron" boundary, since setting the permeability of unsaturated iron to infinity is a good approximation in most cases). This configuration can be used to approximately model magnets with cylindrical iron yokes surrounding the windings. Solutions for monopole layers (charge sheets) with a closed cylindrical boundaries will not be developed because monopole layers are used only in applying the Helmholtz decomposition, in which the fields are computed with open boundaries. Since only real surface currents are treated in this section, an explicit factor of $\mu_0 NI$ is included in the expressions for potentials, fields, and gradients. With this factor in the front of the expressions, the stream function $f_m(z) \sin m\phi$ is dimensionless and has a peak value of unity (see Section 6, Eq. 34). Therefore $C_m(k)$ has dimensions of length in this section.

The boundary condition for an infinitely permeable surface is $V_{\text{scalar}} = 0$. In order to satisfy both the jump condition at $r = a$ and the zero condition at $r = b$, the radial factor in the scalar potential must be $I_m(kr)$ for $r < a$ and a linear combination of $I_m(kr)$ and $K_m(kr)$ for $a < r < b$. The Bessel-function solution that satisfies these requirements is

$$\begin{aligned} U_m(r, z) &= -\mu_0 NI a \int_0^\infty k I_m(kr) \left[K'_m(ka) - \frac{K_m(kb) I'_m(ka)}{I_m(kb)} \right] C_m(k) \cos kz dk, \quad r < a \\ U_m(r, z) &= -\mu_0 NI a \int_0^\infty k I'_m(ka) \left[K_m(kr) - \frac{K_m(kb) I_m(kr)}{I_m(kb)} \right] C_m(k) \cos kz dk, \quad a < r < b \end{aligned} \quad (132)$$

To get the on-axis gradient, we follow the same steps used in deriving Eq. 123, but add the iron term to the Bessel-function kernel and put in the $\mu_0 NI$ factor. This gives

$$g_m(z) = -\mu_0 NI a \frac{1}{2^{m-1}(m-1)!} \int_0^\infty k^{m+1} \left[K'_m(ka) - \frac{K_m(kb) I'_m(ka)}{I_m(kb)} \right] C_m(k) \cos kz dk \quad (133)$$

We note that in the limit of large b , without the $\mu_0 NI$ factor, the potential of Eq. 132 approaches the open-boundary result of Eq. 121. The other limiting case of interest is $b = a$. This limit is approximated by current-sheet magnets wrapped with permeable material, commonly used as steering magnets or correctors. In this limit the scalar potential is

$$U_m(r, z) = \mu_0 NI \int_0^\infty \frac{I_m(kr)}{I_m(ka)} C_m(k) \cos kz dk \quad (134)$$

Following the steps of the previous section, we get for the gradient-length product gL_{eff} for the current-sheet multipole magnet with winding radius a and iron radius b

$$gL_{\text{eff}} = \mu_0 NI \frac{m}{2a^m} \left[1 + \left(\frac{a}{b} \right)^{2m} \right] \int_{-\infty}^{+\infty} f_m(z) dz \quad (135)$$

Comparison of Eqs. 131 and 135 shows that for the same current distribution, the iron boundary increases the gradient-length product by a factor of $1 + (a/b)^{2m}$.

12.7 Relation between the potentials and gradients of surface charge distributions and radially pointing dipole distributions

The reader may have already noticed the connection between the scalar potentials of charge sheets and radially-pointing dipole sheets (double layers). Except for the leading factor of a in the expressions, which comes from the area element $ad\phi dz'$ on the cylinder, and is not differentiated, the scalar potential for a unit-density double layer is obtained by applying the operator $-\partial/\partial a$ to the unit-density single-layer (charge-sheet) potential. For open boundary-solutions in the form of integrals over angle and z , one can compare Eq. 9 with Eq. 38. For Bessel-function integral solutions for open boundaries, one can compare Eq. 112 with Eq. 121. In the case of Bessel-function integral solutions for closed boundaries (infinitely permeable cylinder at radius b), one can compare Eq. 110 with Eq. 132. Not surprisingly, the same relation holds also for the on-axis gradient. In the open-boundary case one can compare Eq. 20 with Eq. 50, or Eq. 97 with Eq. 123. In the closed-boundary case one can compare Eq. 113 with Eq. 133.

The relation between the single and double-layer potentials in our cylindrical geometry is a particular example of a more general relation in potential theory, as shown in Appendix E.

12.8 Effective sources for the iron contribution to the field of a distribution of radially-pointing point magnetic dipoles with a closed cylindrical boundary

The second term in the integrands of Eqs. 132 gives the contribution of the iron to the field. Now suppose we had a magnetic-charge distribution on a cylinder of radius b with Fourier transform $S_m^{\text{iron}}(k)$. Then, according to Eq. 112, the open-boundary scalar potential for $r < b$ due to this distribution would be

$$U_m(r, z) = b \int_0^\infty I_m(kr) K_m(kb) S_m^{\text{iron}}(k) \cos kz dk, \quad (136)$$

Then, in order for the iron contribution to the field for $r < a$ from Eq. 136 to agree with that of Eqs. 132, we must have

$$S_m^{\text{iron}(k)} = \mu_0 N I \frac{a k}{b} \frac{I'_m(ka)}{I_m(kb)} C_m(k) \quad (137)$$

The Fourier back transform of $S_m^{\text{iron}}(k)$ gives

$$\sigma_m^{\text{iron}}(z) = \mu_0 N I \frac{a}{b} \int_0^\infty k \frac{I'_m(ka)}{I_m(kb)} C_m(k) \cos kz dk \quad (138)$$

As k approaches zero, the Bessel-function factor in Eq. 138 approaches a finite limit:

$$\lim_{k \rightarrow 0} \frac{a k}{b} \frac{I'_m(ka)}{I_m(kb)} = \frac{m}{b} \left(\frac{a}{b} \right)^m \quad (139)$$

The scalar potential and field components for $r < a$ for a current-sheet source on a cylinder of radius a inside of a cylindrical iron boundary of radius b are therefore replicated by the sum of the open-boundary potential and fields from the double distribution $f_m(z) \sin m\phi$ on the cylinder of radius a and those of a charge distribution $\sigma_m^{\text{iron}}(z) \sin m\phi$ on a cylinder of radius b . This means that the on-axis gradient can be computed from the integrals of Eqs. 20 and 50. That is,

$$g_m(z) = -\mu_0 N I a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} f_m(z') \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z - z')^2]^{m+1/2}} dz' + \frac{b^{m+1}(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} \frac{\sigma_m^{\text{iron}}(z')}{[b^2 + (z - z')^2]^{m+1/2}} dz' \quad (140)$$

In order to reduce computation time, σ_m^{iron} can be precomputed by Filon numerical integration on a finely-spaced set of z points and stored. Values of $d\sigma_m^{\text{iron}}/dz$ can also be precomputed by differentiation of the right-hand side of Eq. 138 inside of the k integral, which gives a factor of $-k \sin kz$ in place of $\cos kz$. The derivative values are evaluated and stored at the same grid points to facilitate construction of a cubic interpolant.

An alternative to replicating the iron contribution with a charge sheet of radius b is to use a current sheet (double layer) with radius a . This approach was used in the cylindrical magnet models in the existing version of Marylie. In this approach, the double-layer density representing the iron contribution is

$$f_m^{\text{iron}}(z) = - \int_0^\infty \frac{I'_m(ka) K_m(kb)}{I_m(kb) K'_m(ka)} C_m(k) \cos kz dk \quad (141)$$

This Bessel-function factor also approaches a finite limit as k approaches zero:

$$\lim_{k \rightarrow 0} - \frac{I'_m(ka) K_m(kb)}{I_m(kb) K'_m(ka)} = \left(\frac{a}{b} \right)^{2m} \quad (142)$$

Then the gradient including the effect of the iron boundary is just

$$g_m(z) = -a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} [f_m(z') + f_m^{\text{iron}}(z')] \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z - z')^2]^{m+1/2}} dz' \quad (143)$$

12.9 Special case: effective sources for the field in the limit $b = a$

The scalar potential for a current sheet with no gap between the current sheet and the iron boundary was given in Eq. 134. The radial field for this limiting case is then

$$B_r(r, \theta, z) = \mu_0 N I \sin m\theta \frac{k I'_m(kr)}{I_m(ka)} C_m(k) \cos kz dk \quad (144)$$

and the azimuthal field is

$$B_\theta(r, \theta, z) = \mu_0 N I \cos m\theta \int_0^\infty \frac{m I_m(kr)}{r I_m(ka)} C_m(k) \cos kz dk \quad (145)$$

We can now apply the Helmholtz decomposition to these field distributions. According to Section 9, Eqs. 75, 76, and 145, the Helmholtz-decomposition stream function (double-layer density) for the $b = a$ case is derived from $B_\theta(a, z)$ and is

$$\psi_m(\theta, z) = \mu_0 N I \sin m\theta \int_0^\infty C_m(k) \cos kz dk = \mu_0 N I \sin m\theta f_m(z) \quad (146)$$

This is a special case of a more general result: the difference in scalar potential between two points on an infinitely permeable surface with a surface current distribution is just the μ_0 times the total current crossing a curve on the surface connecting the two points. This idea is used to construct printed-circuit-type corrector windings that are glued to iron pole faces [13]. The winding pattern is derived from a grid of measurements of the tangential field on the pole-piece surface.

According to the Helmholtz decomposition, the effective magnetic charge distribution is $b_r^m(z)$ (see Section 9, Eq. 70). In this special $b = a$ case

$$\sigma_m^{\text{iron}}(z) = \mu_0 N I \int_0^\infty k \frac{I'_m(ka)}{I_m(ka)} C_m(k) \cos kz dk \quad (147)$$

It is easily shown by use of the Wronskian relation that the sum of the open-boundary scalar potentials from the double-layer distribution of Eq. 146 and the monopole layer of Eq. 147 is equal to the scalar potential of Eq. 134.

13 Numerical example of Helmholtz decomposition using Bessel-function integrals

This section is a description of how magnetic fields were numerically reconstructed inside of a reference cylinder from field components on the cylinder, using a numerical Fourier- integral calculation. This was done in two ways: Subroutines for computing the magnetic field from a cylindrical filament Lambertson-type quadrupole coil (see Fig. 4) were used to generate magnetic-field test data for the Helmholtz decomposition. The coil was built up out of straight filaments and circular arcs without turn-to-turn connections. The coil was designed to have an (almost) pure normal quadrupole z -integrated field, but the three-dimensional field had strong "allowed" harmonics: $m = 6, m = 10$, etc. as a function of z . These harmonics were in fact considerably larger than those of a typical quadrupole magnet and therefore served as a good test of the algorithms when applied to the higher harmonics. The winding cylinder had a radius of 0.1 m; the data cylinder for the Helmholtz decomposition had a radius r_{ref} of 0.08 m.

Fourier angular coefficients for B_r , B_ϕ , and B_z (i.e. b_r^m , b_ϕ^m , and b_z^m) were numerically evaluated for a set of z points at $r = r_{\text{ref}}$ over a range of z that extended to regions of negligible field. Fictitious charge and dipole distributions were derived from b_r^m and b_ϕ^m , as described in Section 9, Eqs. 72 and 76. The z component b_z^m was not used to construct fictitious sources, but was used for checking the reconstructed fields.

13.1 Field reconstruction by use of numerical Fourier transforms with modified Bessel-function kernels

Fields were reconstructed by Fourier transforms with modified-Bessel-function factors as in Section 12. The $b_r^m(z)$ and $b_\phi^m(z)$ coefficients were numerically Fourier-transformed in z as in Eqs. 87 (only the cosine transform was needed because $b_r^m(z)$ and $b_\phi^m(z)$ were symmetric in z) using a Filon integration routine. The transforms were stored as arrays with finely spaced k values over the range of k for which the transforms were non-negligible in magnitude.

A radius of 0.05 m was used for comparison of directly computed and reconstructed fields. The contribution of each fictitious monopole layer $b_r^m(z) \sin m\phi$ with $m = 2, 6$, and 10 to the field was numerically computed, again by a Filon integrator, as Fourier integrals over k according to Eqs. 95, using the cosine transform of b_r^m in place of $S_{\text{even}}^m(k)$. The contribution of the fictitious currents was evaluated by numerically evaluating the integrals in Eqs. 122 using a/m times the cosine transform of b_ϕ^m in place of $C_m(k)$.

In order to compute the total reconstructed fields, the contributions of the monopole layer and double layer to the components are summed. Figures 6a-6f are a comparison of the Fourier $m = 2$ and $m = 6$ summed field components from the Helmholtz reconstruction, with the Fourier field components directly computed with the Lambertson field algorithm. Numerical agreement is very good.

13.2 Field reconstruction by first computing the on-axis gradient and its derivatives from effective charge-sheet and dipole-sheet distributions, and then using the off-axis expansion of Appendix B

The interface of various magnet models with numerical map integration routines of high-order beam-optics codes, such as Marylie, is the the on-axis gradient and its z derivatives. For this purpose, a subroutine package that can compute the the gradient and its derivatives at arbitrary z values is required. In the particular case of

14 References

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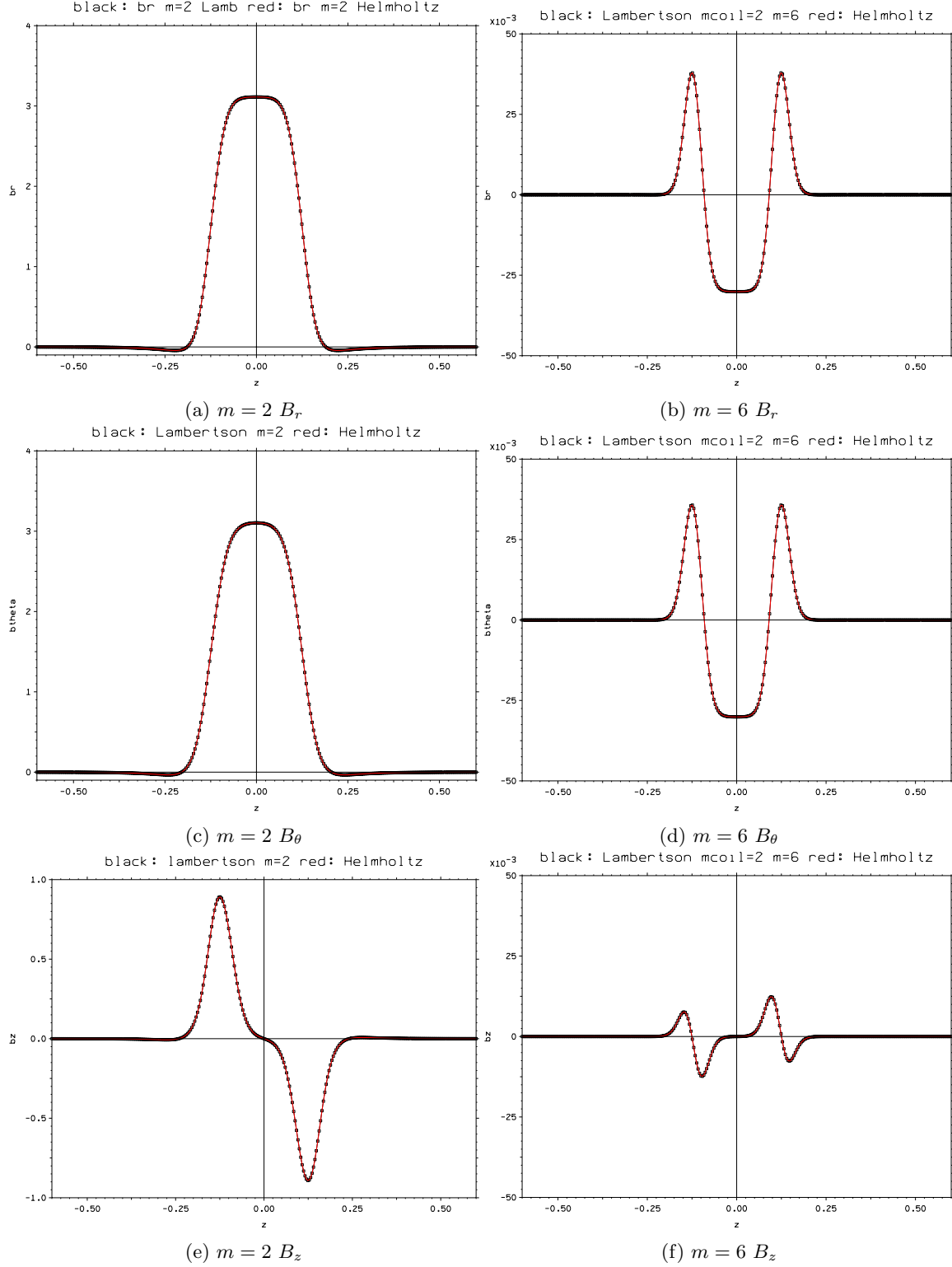


Figure 6: Fourier components of the field from a Lambertson quadrupole. Black squares: direct Biot-Savart filament-model computation; red curves: Helmholtz reconstruction. Lambertson coil radius = 0.1 m, Helmholtz source cylinder radius = 0.08 m, field-point radius = 0.05 m. Fields were reconstructed from Helmholtz sources by use of Fourier transforms with Bessel-function kernels.

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15 Appendix A: Legendre functions used in expressions for the multipole scalar potential in cylindrical coordinates

It is shown here that the function

$$U_Q(r, z) = \frac{1}{\sqrt{ar}} Q_{m-1/2} \left(\frac{a^2 + r^2 + z^2}{2ar} \right) \quad (\text{A1})$$

where $Q_{m-1/2}$ is a Legendre function of the second kind, is a solution of the reduced Laplace equation (Eq. 3). It is convenient to define the variable

$$p = \frac{a^2 + r^2 + z^2}{2ar} \quad (\text{A2})$$

Then U_Q is simply

$$U_Q = \frac{1}{\sqrt{ar}} Q_{m-1/2}(p) \quad (\text{A3})$$

It was shown in Section 2 that except for a multiplicative constant, the potential of Eq. A1 is the reduced scalar potential of a source ring of radius a at $z' = 0$ with a line-charge density that varies as $\sin m\phi$.

The Legendre function $Q_\nu(x + iy)$ is defined in the complex plane with a cut from $-\infty$ to 1. The notation Q_ν is used when the superscript μ in an associated Legendre function Q_ν^μ is zero. In standard terminology μ is the order and ν the degree. The function $Q_\nu(x + iy)$ has branch points at -1 , $+1$, and, if ν is not an integer, at $-\infty$. For our potential problems, the argument p is real and lies on the positive real axis between $+1$ and $+\infty$, and $\nu = m - 1/2$.[†] The function has a logarithmic singularity as p approaches 1 from above. Physically, this means that the field point is approaching the ring. This is analogous to the logarithmic singularity in the scalar potential for a line charge as the field point approaches the line. As r approaches zero, p approaches infinity and $Q_{m-1/2}(p)$ approaches zero in such a way that $Q_{m-1/2}(p)/\sqrt{r}$ is proportional to r^m (see Section 3).

It should also be noted that since U_Q is a solution of the reduced Laplace equation (Eq. 3), its partial derivative with respect to the parameter a is also a solution. This fact comes into play when the scalar potential of a ring with a sinusoidally-varying radially-pointing line dipole density to be found.

Also, there is a close relation between $Q_{m-1/2}$ and the complete elliptic integrals E and K (see, for example, Ref. [1], **8.13.3** and **8.13.7**, p. 337). This connection is apparent when one of Kummer's quadratic transformations (published in 1834) is applied to the hypergeometric series for $Q_{m-1/2}$ (Eq. 14). For example, when $m = 0$, $\nu = -1/2$, and Eq. 14 becomes

$$Q_{-1/2}(p) = \frac{\Gamma^2(1/2)}{\sqrt{2}\Gamma(1)} \left(\frac{1}{p}\right)^{1/2} F\left(\frac{3}{4}, \frac{1}{4}, 1; \frac{1}{p^2}\right) = \frac{\pi}{\sqrt{2}} \left(\frac{1}{p}\right)^{1/2} F\left(\frac{3}{4}, \frac{1}{4}, 1; \frac{1}{p^2}\right) \quad (\text{A4})$$

Using the fact that F is unchanged when its first two parameters are exchanged, and using the transformation (see Ref. [1], p. 560, **15.3.20**)

$$F(a, a + 1/2, c; z) = \left(\frac{1}{1 + \sqrt{z}}\right)^{2a} F\left(2a, c - 1/2, 2c - 1; \frac{2\sqrt{z}}{1 + \sqrt{z}}\right) \quad (\text{A5})$$

Eq. A4 becomes

$$Q_{-1/2}(p) = \frac{\pi}{2} k F\left(\frac{1}{2}, \frac{1}{2}, 1; k^2\right) \quad (\text{A6})$$

with

$$k^2 = \frac{4ar}{(a + r)^2 + z^2} \quad (\text{A7})$$

Using the hypergeometric series for the complete elliptic integral $K(k)$ (see Ref. [1], p. 591, **17.3.9**)

$$K(k) = \frac{\pi}{2} F\left(\frac{1}{2}, \frac{1}{2}, 1; k^2\right) \quad (\text{A8})$$

we get

$$Q_{-1/2}(p) = kK(k) \quad (\text{A9})$$

[†]Note: Computer algorithms are sometimes given for the function $Q_\nu(x)$ on the interval $[-1, +1]$. This function is the average of the values for $Q_\nu(x + iy)$ on either side of the cut, i.e.

$$Q_\nu(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{2} [Q_\nu(x + i\epsilon) + Q_\nu(x - i\epsilon)]$$

The above function is *not* to be confused with the function Q_ν defined on the interval $[1, +\infty]$. The function on the real interval $[1, +\infty]$ is the one of present interest.

A similar procedure using in addition to Eq. A8 the relation for the elliptic integral $E(k) = \frac{\pi}{2}F\left(-\frac{1}{2}, \frac{1}{2}, 1; k^2\right)$ (see Ref. [1], p. 591, **17.3.10**) gives the result of Ref. [1], p. 337, **8.13.7**.

$$Q_{1/2}(p) = p\sqrt{\frac{2}{p+1}}K\left(\sqrt{\frac{2}{p+1}}\right) - [2(p+1)]^{1/2}E\left(\sqrt{\frac{2}{p+1}}\right) \quad (\text{A10})$$

When expressed in terms of k , Eq. A10 takes the form

$$Q_{1/2}(p) = \left(\frac{2}{k} - k\right)K(k) - \frac{2}{k}E(k) \quad (\text{A11})$$

Using the elliptic integral $D(k) \equiv [K(k) - E(k)]/k^2$, the final result for $Q_{1/2}$ in terms of elliptic integrals is

$$Q_{1/2}(p) = 2kD(k) - kK(k) \quad (\text{A12})$$

The elliptic-integral expressions for $Q_{-1/2}$ and $Q_{1/2}$ can be used as starting values for computing $Q_{m-1/2}$ for larger m values by recursion using the relation (see Ref. [1], **8.5.3**, p. 334)

$$Q_{m+1/2}(p) = \frac{1}{m+1/2} [2mQ_{m-1/2}(p) - (m-1/2)Q_{m-3/2}(p)] \quad (\text{A13})$$

This recursion scheme gives good precision if m is not too large ($m < 11$ and p is not too large. For large p , the hypergeometric series form gives better accuracy.

15.1 Proof that $U(r, z)$ of Eq. A1 is a solution of the Laplace equation

The Legendre function $Q_\nu(p)$ satisfies the differential equation (see Ref. [1], **8.1.1**, p. 332, with $\mu = 0$)

$$(1-p^2)\frac{d^2Q_\nu}{dp^2} - 2p\frac{dQ_\nu}{dp} + \nu(\nu+1)Q_\nu = 0 \quad (\text{A14})$$

With $\nu = m - 1/2$, this becomes

$$(1-p^2)\frac{d^2Q_{m-1/2}}{dp^2} - 2p\frac{dQ_{m-1/2}}{dp} + (m^2 - \frac{1}{4})Q_{m-1/2} = 0 \quad (\text{A15})$$

Substituting U of Eq. A1 into the reduced Laplace equation (Eq. 3) and multiplying through by $\sqrt{ar}$ gives

$$\left[\left(\frac{\partial p}{\partial r}\right)^2 + \left(\frac{\partial p}{\partial z}\right)^2\right]\frac{d^2Q_{m-1/2}}{dp^2} + \left[\frac{\partial^2 p}{\partial r^2} + \frac{\partial^2 p}{\partial z^2}\right]\frac{dQ_{m-1/2}}{dp} + \frac{\frac{1}{4} - m^2}{r^2}Q_{m-1/2} = 0 \quad (\text{A16})$$

Next, we use the definition of p (see Eq. A1) to find that

$$\left(\frac{\partial p}{\partial r}\right)^2 + \left(\frac{\partial p}{\partial z}\right)^2 = \frac{p^2 - 1}{r^2} \quad \text{and} \quad \frac{\partial^2 p}{\partial r^2} + \frac{\partial^2 p}{\partial z^2} = \frac{2p}{r^2} \quad (\text{A17})$$

Putting these expressions into Eq. A16 and multiplying through by r^2 gives Eq. A15, which completes the proof.

16 Appendix B: Expansion of a pure multipole scalar potential from the on-axis generalized gradient and its derivatives

As in Sections 3 and 6, I define the on-axis generalized gradient to be

$$g_m(z - z') = m \lim_{r \rightarrow 0} \frac{U_m(r, z - z')}{r^m} \quad (\text{B1})$$

According to Whittaker and Watson (Ref. [4]), Chapter 18, MISCELLANEOUS EXAMPLES, 2., p. 400, if the scalar potential near the axis approaches the limiting form $V(r, \theta, z) = r^m \sin m\theta g_m(z)$ (in my notation and with the factor of m in g_m as in Eq. B12) as r approaches zero, then the off-axis reduced potential U_m is given by the expression

$$U_m(r, z) = \frac{(m-1)!r^m}{\Gamma(m + \frac{1}{2})\Gamma(\frac{1}{2})} \int_0^\pi g_m(z + ir \cos t) \sin^{2m} t dt, \quad (\text{B2})$$

where Γ is the usual gamma function. Substituting the expressions for the gamma function of half-integer argument, we get

$$U_m(r, z) = \frac{(m-1)!r^m 2^m}{\pi(2m-1)!!} \int_0^\pi g_m(z + ir \cos t) \sin^{2m} t dt, \quad (\text{B3})$$

Eq. B2 can be regarded as a 3-D analog of analytic continuation in 2-D potential theory.

At first glance it may appear that the potential function defined by Eq. B2 is complex, but a Taylor expansion of $g_m(z + ir \cos t)$ in powers of r gives

$$\begin{aligned} g_m(z + ir \cos t) = g_m(z) + ir \cos t \frac{d}{dz} g_m(z) + \frac{(ir)^2}{2} \cos^2 t \frac{d^2}{dz^2} g_m(z) + \frac{(ir)^3}{6} \cos^3 t \frac{d^3}{dz^3} g_m(z) + \dots \\ + \frac{(ir)^l}{l!} \cos^l t \frac{d^l}{dz^l} g_m(z) \dots \end{aligned} \quad (\text{B4})$$

We now note that since for odd l , $\cos^l t$ is an odd function around $\pi/2$, and $\sin^{2m} t$ is even, when the Taylor expansion is substituted for $g_m(z + i \cos t)$ in Eq. B2, all of the the terms with odd l integrate to zero, and the integral can be written in the form

$$\int_0^\pi g_m(z + ir \cos t) \sin^{2m} t dt = \sum_{n=0}^{\infty} \left[\frac{1}{(2n)!} \left(-r^2 \frac{d^2}{dz^2} \right)^n g_m(z) \int_0^\pi \cos^{2n} t \sin^{2m} t dt \right] \quad (\text{B5})$$

According to the symbolic mathematics program **Maxima**

$$\int_0^\pi \cos^{2n} t \sin^{2m} t dt = B(m + 1/2, n + 1/2) \quad (\text{B6})$$

where B is the beta function. Using the expression for the beta function in terms of the gamma function, we get

$$\int_0^\pi \cos^{2n} t \sin^{2m} t dt = \frac{\Gamma(m + \frac{1}{2}) \Gamma(n + \frac{1}{2})}{\Gamma(m + n + 1)} = \frac{\pi}{2^{m+n}} \frac{(2m-1)!!(2n-1)!!}{(n+m)!} \quad (\text{B7})$$

Using the identity

$$\frac{(2n-1)!!}{(2n)!} = \frac{1}{2^n n!} \quad (\text{B8})$$

and multiplying all of the factors out, we get the well-known final result

$$U_m(r, z) = r^m (m-1)! \sum_{n=0}^{\infty} \frac{1}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n g_m(z) \quad (\text{B9})$$

If $g_m(z)$ is derived from surface sources on a cylinder of radius a with analytic solutions to the Laplace equation, Eq. B9 is valid for a finite range of $r < r_{\min}$. For source distributions bounded in z , $r_{\min} = a$ for z inside of the source distribution; for z outside of the source distribution, $r_{\min} = (a^2 + z_{\min}^2)^{1/2}$, where $z_{\min}$ is the z distance to the nearest source point.

It should be pointed out that if $g_m(z)$ is an arbitrary function used to fit data, Eq. B9 can have a much smaller range of validity, and may not properly represent fields off axis, especially near sources.

The field components are obtained by taking the gradient of $\sin m\theta U_m$:

$$\begin{aligned} B_r &= \sin m\theta r^{m-1} (m-1)! \sum_{n=0}^{\infty} \frac{m+2n}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n g_m(z) \\ B_\theta &= \cos m\theta r^{m-1} m! \sum_{n=0}^{\infty} \frac{1}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n g_m(z) \\ B_z &= \sin m\theta r^m (m-1)! \sum_{n=0}^{\infty} \frac{1}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n \frac{d}{dz} g_m(z) \end{aligned} \quad (\text{B10})$$

The vector potential in the gauge with $A_\theta = 0$ is obtained by use of Eqs. 69 of Section 8:

$$\begin{aligned} A_r &= \cos m\theta \frac{r}{m} \frac{\partial U_m}{\partial z} = \cos m\theta r^{m+1} \frac{(m-1)!}{m} \sum_{n=0}^{\infty} \frac{1}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n \frac{d}{dz} g_m(z) \\ A_\theta &= 0 \\ A_z &= -\cos m\theta \frac{r}{m} \frac{\partial U_m}{\partial r} = -\cos m\theta r^m \frac{(m-1)!}{m} \sum_{n=0}^{\infty} \frac{m+2n}{(n+m)!n!} \left(-\frac{r^2}{4} \frac{d^2}{dz^2} \right)^n g_m(z) \end{aligned} \quad (\text{B11})$$

16.1 Alternative derivation of the series expansion for U_m

The following section gives the more conventional derivation of the series expansion for $U_m(r, z)$ in terms of the on-axis gradient and its derivatives. As before, I define the on-axis generalized gradient to be

$$g_m(z - z') = m \lim_{r \rightarrow 0} \frac{U_m(r, z - z')}{r^m} \quad (\text{B12})$$

The reduced Laplace equation in cylindrical coordinates for a pure m -pole was given earlier and is repeated below:

$$\frac{\partial^2 U_m}{\partial r^2} + \frac{1}{r} \frac{\partial U_m}{\partial r} + \frac{\partial^2 U_m}{\partial z^2} - \frac{m^2}{r^2} U_m = 0 \quad (\text{B13})$$

We now postulate a solution of the form

$$U_m(r, z) = \sum_{n=0}^{\infty} f_{m,n}(z) r^{m+2n} \quad (\text{B14})$$

When this series is inserted into Eq. B13 we get

$$\sum_{n=0}^{\infty} \left[4n(m+n) r^{m+2n-2} f_{m,n}(z) + r^{m+2n} \frac{d^2}{dz^2} f_{m,n}(z) \right] = 0 \quad (\text{B15})$$

In order for Eq. B15 to be satisfied for arbitrary r , the factors multiplying the same powers of r must individually vanish. This gives the relation

$$f_{m,n}(z) = -\frac{1}{n(m+n)} \frac{1}{4} \frac{d^2}{dz^2} f_{m,n-1}(z) \quad (\text{B16})$$

Next, starting with $n = 1$ and iterating, we have the relations

$$\begin{aligned} f_{m,1}(z) &= \frac{1}{m+1} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right) f_{m,0}(z) \\ f_{m,2}(z) &= \frac{1}{2(m+2)(m+1)} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right)^2 f_{m,0}(z) \\ f_{m,3}(z) &= \frac{1}{3 \cdot 2(m+3)(m+2)(m+1)} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right)^3 f_{m,0}(z) \\ &\quad \dots \\ f_{m,n}(z) &= \frac{1}{n!(m+n)(m+n-1) \dots (m+3)(m+2)(m+1)} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right)^n f_{m,0}(z) \end{aligned} \quad (\text{B17})$$

and

$$f_{m,n}(z) = \frac{m!}{n!(m+n)!} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right)^n f_{m,0}(z) \quad (\text{B18})$$

With the definition of Eq. B12 for $g_m(z)$, this gives $f_{m,0}(z) = (1/m)g_m(z)$ and the final result

$$U_m(r, z) = r^m \sum_{n=0}^{\infty} \frac{(m-1)!}{n!(m+n)!} \left(-\frac{1}{4} \frac{d^2}{dz^2} \right)^n g_m(z), \quad (\text{B19})$$

which agrees with Eq. B9.

17 Appendix C: On-axis generalized gradient and its derivatives for a surface-charge distribution represented by a piecewise-continuous polynomial

Using the variable x for the source z value and z for the field point z , we have for a bounded charge distribution $f_m(x)$ with particular multipole number m

$$g_m(z) = \frac{a^{m+1}(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} \frac{f_m(x)}{[a^2 + (z-x)^2]^{m+1/2}} dx = C_m a^{m+1} \int_{-\infty}^{+\infty} \frac{f_m(x)}{[a^2 + (z-x)^2]^{m+1/2}} dx, \quad (C1)$$

which defines C_m . We now divide the source region into a set of intervals $[x_0, x_1], [x_1, x_2], \dots, [x_i, x_{i+1}], \dots, [x_N, x_{N+1}]$ and assume that the source distribution $f_m(x)$ is approximated by a set of polynomials of up to 4th degree on the intervals. On a typical interval $[x_i, x_{i+1}]$, we set $f_m(x) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4$ with $t = x - x_i$. We have for the contribution to g_m from this interval

$$\Delta g_m(z) = C_m a^{m+1} \int_0^\Delta \frac{\alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4}{[a^2 + (z - x_i - t)^2]^{m+1/2}} dt = C_m a^{m+1} I_m \quad (C2)$$

with $\Delta = x_{i+1} - x_i$ and

$$I_m = \int_0^\Delta \frac{\alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 t^4}{[a^2 + (z - x_1 - t)^2]^{m+1/2}} dt \quad (C3)$$

We now set $s = x_1 + t - z$, $s_1 = x_1 - z$, and $s_2 = x_1 + \Delta - z$. Then $t = s + z - x_1$, or, defining $z_1 = z - x_1$, $t = s + z_1$. The integral over the interval now takes the form

$$\begin{aligned} \int_0^\Delta \frac{\alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3}{[a^2 + (z - x_1 - t)^2]^{m+1/2}} dt &= \int_{s_1}^{s_2} \frac{\alpha_0 + \alpha_1(s + z_1) + \alpha_2(s + z_1)^2 + \alpha_3(s + z_1)^3 + \alpha_4(s + z_1)^4}{[a^2 + s^2]^{m+1/2}} ds \\ &= \int_{s_1}^{s_2} \frac{A_0 + A_1 s + A_2 s^2 + A_3 s^3 + A_4 s^4}{[a^2 + s^2]^{m+1/2}} ds \end{aligned} \quad (C4)$$

with

$$\begin{aligned} A_0 &= \alpha_0 + \alpha_1 z_1 + \alpha_2 z_1^2 + \alpha_3 z_1^3 + \alpha_4 z_1^4 & A_1 &= \alpha_1 + 2\alpha_2 z_1 + 3\alpha_3 z_1^2 + 4\alpha_4 z_1^3 \\ A_2 &= \alpha_2 + 3\alpha_3 z_1 + 6\alpha_4 z_1^2 & A_3 &= \alpha_3 + 4\alpha_4 z_1 & A_4 &= \alpha_4 \end{aligned} \quad (C5)$$

We will also need the z derivatives of $A_0 - A_4$ in calculation of the z derivatives of I_m :

$$\begin{aligned} \frac{\partial A_0}{\partial z} &= \alpha_1 + 2\alpha_2 z_1 + 3\alpha_3 z_1^2 + 4\alpha_4 z_1^3 & \frac{\partial A_1}{\partial z} &= 2\alpha_2 + 6\alpha_3 z_1 + 12\alpha_4 z_1^2 \\ \frac{\partial A_2}{\partial z} &= 3\alpha_3 + 12\alpha_4 z_1 & \frac{\partial A_3}{\partial z} &= 4\alpha_4 & \frac{\partial A_4}{\partial z} &= 0 \end{aligned} \quad (C6)$$

$$\begin{aligned} \frac{\partial^2 A_0}{\partial z^2} &= 2\alpha_2 + 6\alpha_3 z_1 + 12\alpha_4 z_1^2 & \frac{\partial^2 A_1}{\partial z^2} &= 6\alpha_3 + 24\alpha_4 z_1 \\ \frac{\partial^2 A_2}{\partial z^2} &= 12\alpha_4 & \frac{\partial^2 A_3}{\partial z^2} &= \frac{\partial^2 A_4}{\partial z^2} = 0 \end{aligned} \quad (C7)$$

$$\frac{\partial^3 A_0}{\partial z^3} = 6\alpha_3 + 24\alpha_4 z_1 \quad \frac{\partial^3 A_1}{\partial z^3} = 24\alpha_4 \quad \frac{\partial^3 A_2}{\partial z^3} = \frac{\partial^3 A_3}{\partial z^3} = \frac{\partial^3 A_4}{\partial z^3} = 0 \quad (\text{C8})$$

$$\frac{\partial^4 A_0}{\partial z^4} = 24\alpha_4 \quad \frac{\partial^4 A_1}{\partial z^4} = \frac{\partial^4 A_2}{\partial z^4} = \frac{\partial^4 A_3}{\partial z^4} = \frac{\partial^4 A_4}{\partial z^4} = 0 \quad (\text{C9})$$

All higher z derivatives of A_1 to A_4 are zero.

Expressions for the s^n integrals

$$\int \frac{s^n}{(a^2 + s^2)^{m+1/2}} ds, \quad n = 0, 1, 2, 3, 4, m = 1, 2, 3, \dots \quad (\text{C10})$$

are given in Gradstein and Ryzhik, 2.271, 2.272, and 2.273, pp. 86-87. We now introduce the notation

$$I_n^m = \int_{s_1}^{s_2} \frac{s^n}{(a^2 + s^2)^{m+1/2}} ds, \quad n = 0, 1, 2, 3, 4, m = 1, 2, 3, \dots \quad (\text{C11})$$

We now can write $\Delta g_m(z) = C_m a^{m+1} I_m$ in terms of the above integrals over s instead of the integral over t as follows:

$$I_m = A_0 I_0^m + A_1 I_1^m + A_2 I_2^m + A_3 I_3^m + A_4 I_4^m \quad (\text{C12})$$

17.1 z derivatives of the on-axis generalized gradient for an interval in z

In order to get the z derivatives of $\Delta g_m(z)$, we must differentiate I_m with respect to the field point variable z , and then multiply the derivatives by $C_m a^{m+1}$. The expression for the first derivative has terms from differentiation of the limits of integration $s_1 = x_1 - z$ and $s_2 = x_2 - z$, and from differentiation of the coefficients A_1 - A_4 in the integrand. The result is

$$\begin{aligned} \frac{d}{dz} I_m(z) &= \frac{f_m(x_1)}{(a^2 + s_1^2)^{m+1/2}} - \frac{f_m(x_2)}{(a^2 + s_2^2)^{m+1/2}} + \int_{s_1}^{s_2} \frac{\frac{\partial A_0}{\partial z} + \frac{\partial A_1}{\partial z} s + \frac{\partial A_2}{\partial z} s^2 + \frac{\partial A_3}{\partial z} s^3}{(a^2 + s^2)^{m+1/2}} ds = \\ &= \frac{\alpha_0}{(a^2 + s_1^2)^{m+1/2}} - \frac{\alpha_0 + \alpha_1 \Delta + \alpha_2 \Delta^2 + \alpha_3 \Delta^3 + \alpha_4 \Delta^4}{(a^2 + s_2^2)^{m+1/2}} + \frac{\partial A_0}{\partial z} I_0^m + \frac{\partial A_1}{\partial z} I_1^m + \frac{\partial A_2}{\partial z} I_2^m + \frac{\partial A_3}{\partial z} I_3^m \end{aligned} \quad (\text{C13})$$

For the second derivative, the result is

$$\begin{aligned} \frac{d^2}{dz^2} I_m(z) &= (2m+1) \left[\frac{f_m(x_1)}{(a^2 + s_1^2)^{m+3/2}} - \frac{f_m(x_2)}{(a^2 + s_2^2)^{m+3/2}} \right] + \\ &+ \frac{f'_m(x_1)}{(a^2 + s_1^2)^{m+1/2}} - \frac{f'_m(x_2)}{(a^2 + s_2^2)^{m+1/2}} + \int_{s_1}^{s_2} \frac{\frac{\partial^2 A_0}{\partial z^2} + \frac{\partial^2 A_1}{\partial z^2} s + \frac{\partial^2 A_2}{\partial z^2} s^2}{(a^2 + s^2)^{m+1/2}} ds = \\ &= (2m+1) \left[\frac{\alpha_0}{(a^2 + s_1^2)^{m+3/2}} - \frac{\alpha_0 + \alpha_1 \Delta + \alpha_2 \Delta^2 + \alpha_3 \Delta^3 + \alpha_4 \Delta^4}{(a^2 + s_2^2)^{m+3/2}} \right] + \\ &+ \frac{\alpha_1}{(a^2 + s_1^2)^{m+1/2}} - \frac{\alpha_1 + 2\alpha_2 \Delta + 3\alpha_3 \Delta^2 + 4\alpha_4 \Delta^3}{(a^2 + s_2^2)^{m+1/2}} + \frac{\partial^2 A_0}{\partial z^2} I_0^m + \frac{\partial^2 A_1}{\partial z^2} I_1^m + \frac{\partial^2 A_2}{\partial z^2} I_2^m \end{aligned} \quad (\text{C14})$$

The derivatives can be expressed in a more concise form by defining the following function:

$$D_n^m(s) = \frac{d^n}{ds^n} \frac{1}{(a^2 + s^2)^{m+1/2}} \quad (\text{C15})$$

Then, with

$$\begin{aligned} f_m(x_1) &= \alpha_0 & f_m(x_2) &= \alpha_0 + \alpha_1 \Delta + \alpha_2 \Delta^2 + \alpha_3 \Delta^3 + \alpha_4 \Delta^4 \\ f'_m(x_1) &= \alpha_1 & f'_m(x_2) &= \alpha_1 + 2\alpha_2 \Delta + 3\alpha_3 \Delta^2 + 4\alpha_4 \Delta^3 \\ f''_m(x_1) &= 2\alpha_2 & f''_m(x_2) &= 2\alpha_2 + 6\alpha_3 \Delta + 12\alpha_4 \Delta^2 \\ f'''_m(x_1) &= 6\alpha_3 & f'''_m(x_2) &= 6\alpha_3 + 24\alpha_4 \Delta \\ f''''_m(x_1) &= 24\alpha_4 & f''''_m(x_2) &= 24\alpha_4 \end{aligned} \quad (\text{C16})$$

The expressions for the first four derivatives of I_m are

$$\begin{aligned} \frac{d}{dz} I_m(z) &= f_m(x_1) D_0^m(s_1) - f_m(x_2) D_0^m(s_2) + (\alpha_1 + 2\alpha_2 z_1 + 3\alpha_3 z_1^2) I_0^m \\ &\quad + \frac{\partial A_0}{\partial z} I_0^m + \frac{\partial A_1}{\partial z} I_1^m + \frac{\partial A_2}{\partial z} I_2^m + \frac{\partial A_3}{\partial z} I_3^m \end{aligned} \quad (\text{C17})$$

$$\begin{aligned} \frac{d^2}{dz^2} I_m(z) &= -f_m(x_1) D_1^m(s_1) + f_m(x_2) D_1^m(s_2) + f'_m(x_1) D_0^m(s_1) - f'_m(x_2) D_0^m(s_2) + \\ &\quad + \frac{\partial^2 A_0}{\partial z^2} I_0^m + \frac{\partial^2 A_1}{\partial z^2} I_1^m + \frac{\partial^2 A_2}{\partial z^2} I_2^m \end{aligned} \quad (\text{C18})$$

$$\begin{aligned} \frac{d^3}{dz^3} I_m(z) &= f_m(x_1) D_2^m(s_1) - f_m(x_2) D_2^m(s_2) - f'_m(x_1) D_1^m(s_1) + f'_m(x_2) D_1^m(s_2) + \\ &\quad f''_m(x_1) D_0^m(s_1) - f''_m(x_2) D_0^m(s_2) + \frac{\partial^3 A_0}{\partial z^3} I_0^m + \frac{\partial^3 A_1}{\partial z^3} I_1^m \end{aligned} \quad (\text{C19})$$

$$\begin{aligned} \frac{d^4}{dz^4} I_m(z) &= -f_m(x_1) D_3^m(s_1) + f_m(x_2) D_3^m(s_2) + f'_m(x_1) D_2^m(s_1) - f'_m(x_2) D_2^m(s_2) - \\ &\quad f''_m(x_1) D_1^m(s_1) + f''_m(x_2) D_1^m(s_2) + f'''_m(x_1) D_0^m(s_1) - f'''_m(x_2) D_0^m(s_2) + \frac{\partial^4 A_0}{\partial z^4} I_0^m \end{aligned} \quad (\text{C20})$$

Since the highest power of t in $f_m(x)$ is 4, the 5^{th} and higher derivatives contain no s^n integrals:

$$\begin{aligned} \frac{d^n}{dz^n} I_m(z) &= (-1)^{n-1} \{ f_m(x_1) D_{n-1}^m(s_1) - f_m(x_2) D_{n-1}^m(s_2) - f'_m(x_1) D_{n-2}^m(s_1) + \\ &\quad f'_m(x_2) D_{n-2}^m(s_2) + f''_m(x_1) D_{n-3}^m(s_1) - f''_m(x_2) D_{n-3}^m(s_2) - f'''_m(x_1) D_{n-4}^m(s_1) + f'''_m(x_2) D_{n-4}^m(s_2) - \\ &\quad f''''_m [D_{n-5}^m(s_1) - D_{n-5}^m(s_2)] \} \quad n > 5 \end{aligned} \quad (\text{C21})$$

With $p = m + 1/2$, the expressions for the first four $D_n^m(s)$ are:

$$\begin{aligned}
D_0^m(s) &= \frac{1}{(a^2 + s^2)^p} \\
D_1^m(s) &= \frac{-2ps}{(a^2 + s^2)^{p+1}} \\
D_2^m(s) &= \frac{-2p}{(a^2 + s^2)^{p+1}} + \frac{4p(p+1)s^2}{(a^2 + s^2)^{p+2}} \\
D_3^m(s) &= \frac{12p(p+1)s}{(a^2 + s^2)^{p+2}} - \frac{48p(p+1)(p+2)s^2}{(a^2 + s^2)^{p+3}} + \frac{16p(p+1)(p+2)(p+3)s^4}{(a^2 + s^2)^{p+4}} \quad (\text{C22})
\end{aligned}$$

However, the subroutine for computing a sequence of D_n^m values for $n = 0, 1, 2, 3, \dots, n_{\max}$ for arbitrary $n_{\max}$ uses numerical recursion rather than the algebraic expressions of Eq. C22.

18 Appendix D: On-axis generalized gradient and its derivatives for a surface-dipole distribution represented by a piecewise-continuous polynomial

In this section we extend the results of the previous section to distributions of radially-pointing infinitesimal dipoles on a cylindrical surface. Again using the variable x for the source z value and z for the field point z , we have for a particular multipole number m

$$\begin{aligned}
g_m(z) &= -a \frac{(2m-1)!!}{2^{m+1}(m-1)!} \int_{-\infty}^{+\infty} h_m(x) \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z-x)^2]^{m+1/2}} \\
&= -C_m a \int_{-\infty}^{+\infty} h_m(x) \frac{\partial}{\partial a} \frac{a^m}{[a^2 + (z-x)^2]^{m+1/2}} \quad (\text{D1})
\end{aligned}$$

where C_m is the same as in Eq. 20 and $h_m(x)$ is the dipole density. Performing the a derivative explicitly gives two terms:

$$g_m(z) = -C_m a^m \left[m \int_{-\infty}^{+\infty} \frac{h_m(x)}{[a^2 + (x-z)^2]^{m+1/2}} dx - (2m+1)a^2 \int_{-\infty}^{+\infty} \frac{h_m(x)}{[a^2 + (x-z)^2]^{m+3/2}} dx \right] \quad (\text{D2})$$

Working again with a piecewise-continuous quartic with $t = x - x_1$ on a typical interval $[x_1, x_2]$, we set $h_m(x) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4$. In terms of the variables s and z_1 , $h_m = \beta_0 + \beta_1 z_1 + \beta_2 z_1^2 + \beta_3 z_1^3 + \beta_4 z_1^4 + (\beta_1 + 2\beta_2 z_1 + 3\beta_3 z_1^2 + 4\beta_4 z_1^3)s + (\beta_2 + 3\beta_3 z_1 + 6\beta_4 z_1^2)s^2 + (\beta_3 + 4\beta_4 z_1)s^3 + \beta_4 s^4$. We see that the first integral in Eq. 49 is of the same form as the integral in Eq. 20. The second integral has the same s^n integrals as in Eq. C10, but with $m+1$ in place of m . Next, we set

$$J_m^{(1)} = \int_0^\Delta \frac{\beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4}{[a^2 + (z - x_1 - t)^2]^{m+1/2}} dt \quad (\text{D3})$$

and

$$J_m^{(2)} = \int_0^\Delta \frac{\beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4}{[a^2 + (z - x_1 - t)^2]^{m+3/2}} dt \quad (\text{D4})$$

Using the same approach as in the previous sections, we now have for the contribution to the double-layer g_m from the interval $[x_1, x_2]$

$$\Delta g_m(z) = -C_m a^m \left[m J_m^{(1)} - (2m+1) a^2 J_m^{(2)} \right] \quad (\text{D5})$$

with

$$\begin{aligned} J_m^{(1)} = & (\beta_0 + \beta_1 z_1 + \beta_2 z_1^2 + \beta_3 z_1^3 + \beta_4 z_1^4) I_0^m + (\beta_1 + 2\beta_2 z_1 + 3\beta_3 z_1^2 + 4\beta_4 z_1^3) I_1^m + \\ & (\beta_2 + 3\beta_3 z_1 + 6\beta_4 z_1^2) I_2^m + (\beta_3 + 4\beta_4 z_1) I_3^m + \beta_4 I_4^m \end{aligned} \quad (\text{D6})$$

and

$$\begin{aligned} J_m^{(2)} = & (\beta_0 + \beta_1 z_1 + \beta_2 z_1^2 + \beta_3 z_1^3 + \beta_4 z_1^4) I_0^{m+1} + (\beta_1 + 2\beta_2 z_1 + 3\beta_3 z_1^2 + 4\beta_4 z_1^3) I_1^{m+1} + \\ & (\beta_2 + 3\beta_3 z_1 + 6\beta_4 z_1^2) I_2^{m+1} + (\beta_3 + 4\beta_4 z_1) I_3^{m+1} + \beta_4 I_4^{m+1} \end{aligned} \quad (\text{D7})$$

Then, with

$$\begin{aligned} h_m(x_1) &= \beta_0 & h_m(x_2) &= \beta_0 + \beta_1 \Delta + \beta_2 \Delta^2 + \beta_3 \Delta^3 + \beta_4 \Delta^4 \\ h'_m(x_1) &= \beta_1 & h'_m(x_2) &= \beta_1 + 2\beta_2 \Delta + 3\beta_3 \Delta^2 + 4\beta_4 \Delta^3 \\ h''_m(x_1) &= 2\beta_2 & h''_m(x_2) &= 2\beta_2 + 6\beta_3 \Delta + 12\beta_4 \Delta^2 \\ h'''_m(x_1) &= 6\beta_3 & f'''_m(x_1) &= 6\beta_3 + 24\beta_4 \Delta \\ h''''_m(x_1) &= 24\beta_4 & h''''_m(x_1) &= 24\beta_4 \end{aligned} \quad (\text{D8})$$

and the derivatives of $J_m^{(1)}$ are given by those of Eqs. C17, C18, C19, and C21, but with h_m in place of f_m and β_0, β_1 , etc. in place of α_0, α_1 , etc.

19 Appendix E: Relation between the charge-sheet and dipole-sheet (double-layer) potentials

19.1 The charge-sheet potential in an arbitrary geometry

We start with the scalar potential from a point charge q inside any surrounding boundary that allows linear superposition of potentials and fields. This includes open boundaries and arbitrary closed boundaries of infinitely permeable material in the magnetostatic problems, and conducting surfaces in electrostatic problems. Infinitely long cylinders are considered to be closed boundaries. We denote the point-charge potential for a unit charge by $\Phi(\mathbf{r}, \mathbf{r}')$, where $\mathbf{r}$ is the field point and $\mathbf{r}'$ is a point on the source surface S . In the case of open boundaries (omitting factors of μ_0 or ϵ_0) $\Phi(\mathbf{r}, \mathbf{r}')$ takes the well-known form

$$\Phi(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{r}'|} \quad (\text{E1})$$

In the case of closed boundaries, $\Phi(\mathbf{r}, \mathbf{r}')$ is more complicated and usually does not have a simple analytic form. However, for any point $\mathbf{r}$ inside of the boundary, $\Phi(\mathbf{r}, \mathbf{r}')$ can be written as the sum of the open-boundary potential of Eq. E1 plus a difference term. The open-boundary term becomes

infinite as $|\mathbf{r} - \mathbf{r}'|$ approaches zero, and the difference term approaches a finite value. For $\mathbf{r}$ close to an infinitely permeable (or conducting in the case of electrostatics) surrounding boundary, Φ approaches zero.

Next, we consider a surface-charge distribution $\sigma(\mathbf{r}')$ on a surface S inside of the boundary surface. By superposition, the potential of the charge sheet is given by the surface integral

$$V_c(\mathbf{r}) = \int_S \sigma(\mathbf{r}') \Phi(\mathbf{r}, \mathbf{r}') dS' \quad (\text{E2})$$

19.2 Double-layer potential in an arbitrary geometry

We denote the double-layer distribution by $D(\mathbf{r}')$. The double-layer distribution is the surface density of normally-pointing infinitesimal dipoles on the surface. In Ref. [16], the relation between the single-layer and double-layer potentials was derived for open boundaries. Here, we generalize the limiting process of Ref. [16] to include arbitrary closed boundaries.

We first consider the sum of the point-source potentials of equal and opposite charges on surfaces offset from the original source surface by the vectors $\pm \hat{n}(\mathbf{r}')\delta/2$, where $\hat{n}$ is the unit normal to the surface at $\mathbf{r}'$ and δ is a small distance (in differential geometry, these surfaces are called *parallel surfaces*). That is, we consider the quantity $\Delta\Phi = \Phi(\mathbf{r}, \mathbf{r}' + \hat{n}\delta/2) - \Phi(\mathbf{r}, \mathbf{r}' - \hat{n}\delta/2)$.

The potentials of the charges on the offset surfaces can be approximated by the first two terms of a Taylor series: and we get

$$\begin{aligned} \Phi(\mathbf{r}, \mathbf{r}' + \hat{n}\delta/2) &\approx \Phi(\mathbf{r}, \mathbf{r}') + \hat{n} \cdot \nabla' \Phi(\mathbf{r}, \mathbf{r}')\delta/2 \\ \Phi(\mathbf{r}, \mathbf{r}' - \hat{n}\delta/2) &\approx \Phi(\mathbf{r}, \mathbf{r}') - \hat{n} \cdot \nabla' \Phi(\mathbf{r}, \mathbf{r}')\delta/2 \end{aligned} \quad (\text{E3})$$

and

$$\Delta\Phi \approx \frac{\partial\Phi}{\partial n} \delta \quad (\text{E4})$$

where $\partial/\partial n = \hat{n} \cdot \nabla'$. We then consider the double distribution $D(\mathbf{r}')$ to be formed by the two parallel surfaces with surface-charge densities $\pm\sigma'$ that approach infinite values in such a way that as δ approaches zero, $\sigma'\delta$ approaches $D(\mathbf{r}')$. Then the potential for the double layer is

$$V_d(\mathbf{r}) = \int_S D(\mathbf{r}') \frac{\partial\Phi}{\partial n} dS' \quad (\text{E5})$$

The above relation between the charge-sheet potential and the double-layer potential is completely general and applies to any geometry for which linear superposition holds.

19.3 Application to cylindrical geometry

The charge distribution of a point charge at a, ϕ_0, z' can be written in the form

$$\sigma(a, \phi, z) = \frac{1}{a} \delta(z - z') \delta(\phi - \phi_0) \quad (\text{E6})$$

The delta function in ϕ can be represented by a Fourier series in ϕ :

$$\delta(\phi - \phi_0) = \sum_{m=0}^{\infty} c_m \cos m(\phi - \phi_0) \quad (\text{E7})$$

with $c_0 = 1/(2\pi)$ and $c_m = (1/\pi) \cos m(\phi - \phi_0)$, $m > 0$. A point charge can therefore be represented by an infinite series of sinusoidal charge rings plus a ring with constant charge density. Similarly, a point dipole can be represented by an infinite series of sinusoidal dipole rings plus a ring with a constant dipole density. In this memo, only the sinusoidal components are of interest.

The surface charge density of a sinusoidal charge ring with linear charge density $\lambda_m \sin m\phi$ at z' is $\sigma_m(\phi, z) = \lambda_m \sin m\phi \delta(z - z')$. We can now form a double layer from it by the limiting process of Section 17.2. The reduced scalar potential of a single sinusoidal charge ring in cylindrical coordinates with open boundaries was given in Eq. 13. The analogous expression for the ring surrounded by an infinitely permeable boundary at $r = b$ was given in Eq. 110 as a Fourier integral involving modified Bessel functions. For both cases we write the potential in the form $V_m^c(r, \theta, z; a, z') = U_m^c(r, z; a, z') \sin m\theta$. The a and z' indicate that the ring has radius a and has the axial position z' . The operator $\hat{n} \cdot \nabla'$ for this geometry is

$$\hat{a} \cdot \left(\hat{a} \frac{\partial}{\partial a} + \hat{\phi} \frac{1}{a} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) = \frac{\partial}{\partial a} \quad (\text{E8})$$

The scalar potentials of single-multipole charge sheets and radially-pointing dipole sheets (double layers) derived in this memo are indeed related by the operator $\partial/\partial a$. Except for the leading factor of a in the expressions, which comes from the area element $ad\phi dz'$ on the cylinder, and is not differentiated, the scalar potential for a unit-density double layer is obtained by applying the operator $-\partial/\partial a$ to the unit-density single-layer (charge-sheet) potential. The negative sign is required to ensure that for a normal multipole, B_r at the center of the first pole is positive if ψ is positive at the center of the first pole. For open boundary-solutions in the form of integrals over angle and z , one can compare Eq. 9 with Eq. 38. For Bessel-function integral solutions for open boundaries, one can compare Eq. 112 with Eq. 121. In the case of Bessel-function integral solutions for closed boundaries (infinitely permeable cylinder at radius b), one can compare Eq. 110 with Eq. 132. Not surprisingly, the same relation holds also for the on-axis gradient. In the open-boundary case one can compare Eq. 20 with Eq. 50, or the Bessel-function expression of Eq. 97 with the Bessel-function expression Eq. 123. In the closed boundary case, we can compare the Bessel-function expression of Eq. 113 with the Bessel-function expression of Eq. 133.